Sorted Texture-Aware Glance and Gaze Network for Hyperspectral Image Classification With Low Training Samples

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***Abstract*— Hyperspectral images (HSIs) provide a wealth of information surpassing human visual capabilities, enabling precise identification of remote sensing targets. However, it faces significant challenges, including insufficient long-range dependency modeling, difficulties in data collection, and the tendency of models to get trapped in local optima during training. To overcome these obstacles, we present the sorted texture-aware glance and gaze network (ST-GGNet) tailored for HSI classification. First, we propose the glance and gaze attention (GGA) mechanism, which employs feature interaction-based long-term modeling to minimize information loss across spectral bands and focus on critical land cover features within HSI. Subsequently, the sorted texture-aware module (STM) is introduced to deeply mine and efficiently utilizes detailed texture and spectral information, thereby enhancing accuracy even with limited training data. Additionally, we propose the budding growth optimization algorithm, budding growth optimizer (BGO), which integrates a budding growth mechanism to help the model discover better solutions, boosting optimization and classification performance. Experimental evaluations conducted on four public HSI datasets—Pavia University, Salinas, Houston, and WHU-Longkou—demonstrate the superior performance of ST-GGNet compared to nine state-of-the-art (SOTA) classification methods. Specifically, under limited training samples, ST-GGNet achieves overall accuracies (OAs) of 99.42%, 96.88%, 96.86%, and 97.74%; average accuracies (AAs) of 98.90%, 98.01%, 97.07%, and 92.48%; and Kappa coefficients of 99.24%, 96.53%, 96.59%, and 97.03%, respectively. The findings reveal that ST-GGNet not only maintains strong robustness and generalization but also effectively suppresses noise and excels at distinguishing spatially similar adjacent land covers, especially in low-sample scenarios, consistently outperforming existing SOTA methods. We have released our code and models at https://github.com/Pluviophile-sy/ST-GGNet**

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I. Introduction

HYPERSPECTRAL images (HSIs) capture detailed spectral information by measuring the reflective or radiative properties of targets across hundreds of continuous spectral bands [1]. This capability makes it a vital data source in remote sensing [2], offering a level of precision unattainable with traditional RGB imaging, which is limited to three bands (red, green, and blue). With its superior spectral resolution, HSI enhances the identification and classification of ground objects, enabling the differentiation of visually similar materials based on their unique spectral signatures. This rich spectral information gives HSI-based classification a significant advantage in discriminating between objects with subtle variations [3]. Today, HSI classification is widely employed in diverse fields, including environmental conservation [4], mineral mapping [5], urban ecology [6], land use analysis [7], precision agriculture [8], and medical diagnostics [9]. It se

HSI acquires high-resolution data across hundreds of narrow spectral bands, capturing detailed, nonlinear spectral information within consecutive bands, ranging from hundreds to thousands. This ensures that each pixel contains rich spectral features, encompassing multiple ground objects within a single pixel and often exhibiting high interband correlation, significantly exceeding human visual perception [10]. HSI integrates spatial and spectral information into a complex 3-D data cube, providing substantial advantages in feature identification and classification, particularly in distinguishing visually similar but spectrally distinct targets. Although traditional machine learning methods have made progress in remote sensing fields, such as geophysical measurements [57], geological hazards [58], soil analysis [80], land use [59], and wetland management [60] in recent years, the high-dimensional nature of HSI data leads to information redundancy and increased computational burden [61], making the data more susceptible to noise and the challenges of dimensionality [11]. As a result, traditional methods often struggle to efficiently extract meaningful features from HSI data, highlighting the need for advanced techniques to fully leverage its potential features [12].

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By constructing multilayer neural networks, deep learning can simulate the operation of the human brain, automatically learn complex features from massive data, and realize a deep understanding of data and pattern recognition. Such hierarchical feature learning and automatic feature extraction capabilities show strong application potential in image recognition, speech recognition, natural language processing, and other fields. While its application in the remote sensing field is relatively recent [13], deep learning has significantly advanced hyperspectral data processing [14] by enabling automatic feature extraction [15] and attention mechanisms [16]. These capabilities effectively handle the high dimensionality and information redundancy of HSI data, surpassing traditional methods in capturing high-level semantic information [17], [18]. For instance, Wang et al. [19] employed Gramian angular field encoding combined with a U-Net network for precise pixel-level classification of HSI; Pang et al. [20] introduced graph convolutional networks (GCNs) to optimize HSI representations, enhancing classification accuracy; Yu et al. [21] designed an unsupervised framework for HSI restoration. These studies highlight deep learning's ability to efficiently capture both linear and nonlinear features, independently extract spectral and spatial information, and jointly process them for richer, semantically meaningful representations. Deep learning offers clear advantages over traditional approaches in HIS detection and classification, thanks to its flexible feature extraction capabilities. It delivers higher accuracy and efficiency while overcoming technological limitations, making it a powerful tool for advancing the field.

Convolutional neural networks (CNNs) and Transformers have been extensively employed in deep learning models for HSI classification, thanks to their deep architectures that facilitate powerful feature representations [22], [23]. CNNs mitigate the high dimensionality of HSI data by using sequential convolutional layers to extract progressively lower dimensional features, effectively addressing the problem of gradient vanishing and exploding [24]. Additionally, the convolutional structures in CNNs help suppress noise and irrelevant features during feature extraction, demonstrating robust performance against spectral mixing and noise interference [25]. However, the localized nature of convolution operations restricts CNNs from capturing medium to long-range dependencies across spectral bands, limiting the interaction between features [26]. Furthermore, traditional CNNs are mainly designed for 2-D spatial data, which constrains their ability to model the high-dimensional spectral information in HSI data [27]. This limitation makes it difficult to capture the intricate relationships and dependencies among spectral bands [28], especially with limited data, thereby impacting CNN performance in HSI classification, as shown in Fig. 1(a). In contrast, Transformers have recently exhibited unique strengths in HSI classification through their self-attention mechanisms [29]. These mechanisms efficiently model long-range dependencies between spectral bands, capturing a broader range of spectral features and enhancing spatial feature comprehension [30]. This capability allows Transformers to excel in HSI classification by effectively learning and identifying subtle differences between spectral bands, thereby improving classification accuracy [31].

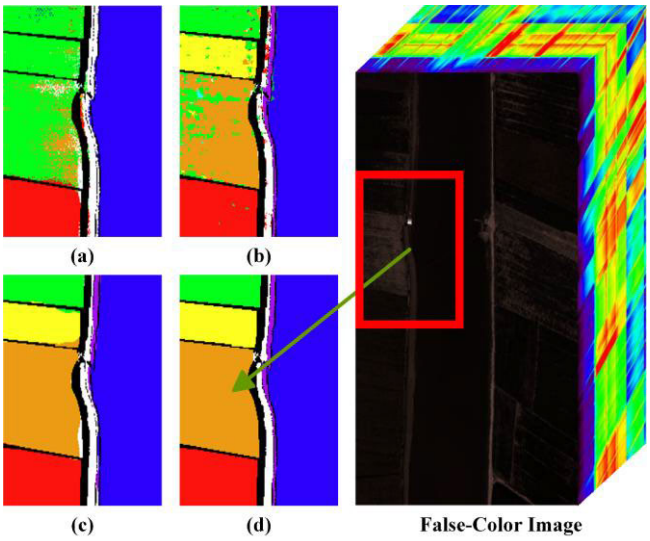


Fig. 1. Illustration of the effectiveness of our method in HSI classification. (a) *CNN-Based:* Owing to inadequate modeling of long-range dependencies and underutilization of spectral information, spatially adjacent ground objects tend to be severely misclassified. (b) *Transformer-Based:* The widespread distribution of boundary noise points, coupled with the inherent complexity of Transformer architectures, can lead to convergence at suboptimal solutions rather than the global optimum. (c) *ST-GGNet:* Relative to (b), its classification accuracy is nearly on par with (a), while in comparison to (c), the resulting map exhibits only a negligible amount of boundary noise. (d) GT.

However, the complex architecture and high number of parameters in Transformer models [32] make them vulnerable to getting stuck in local optima [33] when dealing with the high-dimensional and complex features of HSI data [34], as shown in Fig. 1(b). Recently, Kolmogorov–Arnold networks (KANs) [35] have been introduced, offering superior interpretability, expressive power, and adaptability compared to traditional multilayer perceptions (MLPs) due to the learnable nonlinear activation functions [36]. KANs are gradually being applied in HSI classification for their precise data-fitting capabilities. For example, Jamali et al. [37] developed hybrid models that incorporate 1-D, 2-D, and 3-D KANs, demonstrating enhanced competitiveness in HSI classification tasks. Similarly, Lobanov et al. [38] replaced certain layers in MLP models with KAN layers, achieving significant performance improvements as evidenced by experimental results. However, Jamali et al. [37] conducted experiments only on a self-constructed dataset, without testing their model on other well-established public datasets to assess its performance and generalizability. In the same way, Lobanov et al. [38] employed a straightforward replacement strategy without providing much explanation or analysis. While the application of KAN in HSI classification remains limited, its ability to break down high-dimensional features into univariate functions improves feature representation efficiency and addresses the challenges of dimensionality. This demonstrates considerable potential for further research and development.

Thus, deep learning-based HSI classification must address the following three challenges.

1) *Effective Long-Range Dependency Modeling:* Models often lose critical information when handling long-range dependencies across spectral bands, failing to capture

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subtle interband relationships and adequately focus on key features.

2) *Low Spectral Information Utilization Increasing Sample Dependency:* Current models inadequately extract and utilize spectral information from HSI data, leading to limited spectral feature representation and increased reliance on the number of training samples due to low data utilization efficiency.
3) *High-Dimensional Complexity Leading to Local Optima:* The high-dimensional and complex nature of HSI data makes models susceptible to local optima during training, adversely affecting classification performance.

To enhance long-range dependency modeling, Kurapati and Arun [39] introduced the spatial position encoding (SPE) module, which embeds the spatial position of pixels into spectral data. This equips Transformer models for HSI with a global spatial perspective, allowing them to capture long-range dependencies while preserving spatial consistency. As a result, the architecture design is simplified, and information loss is minimized. However, this method fails to fully exploit the correlation between adjacent spectral bands and lacks an effective strategy for noise handling. Huang et al. [40] developed the spectral–spatial visual foundation model transformer (SS-VFMT), which integrates a visual foundation model with spatial and spectral enhancement modules (SpaEMs and SpeEMs) to better preserve essential features in HSI. They also employed a patch relationship distillation strategy to more effectively utilize pretrained knowledge. However, SPE primarily emphasizes long-range spatial dependencies and has limited ability to model long-term spectral sequences. Additionally, these works focus on local rather than global modeling, leading to insufficient capture of sequential features between spectral bands and potential information loss across different bands.

To maximize the use of spectral information in HSI and mitigate the impact of limited data on classification results, Hong et al. [41] introduced the SpectralGPT model. This model achieves deep integration of spatial and spectral information by employing a 3-D masking strategy to generate spatial–spectral feature representations and a multiobjective reconstruction strategy to capture spectral sequence patterns. These advancements improve the model's adaptability for remote sensing tasks. However, its fusion strategy relies on hard-coded cube patches; the fixed token granularity may lead to information loss or overfusion in HSI scenes with varying scales and structures. Yu et al. [42] proposed HyperSINet, a collaborative interaction network that combines CNN and Transformer to enhance HSI classification performance through the integration of local and nonlocal features. This network includes Conv2Trans and Trans2Conv modules to facilitate feature interaction and transformation between CNN and Transformer branches. However, HyperSINet fails to effectively model spectral similarity, leading to confusion between classes with minor spectral differences and close spatial proximity.

To address the problem of models getting trapped in local optima, various improvements have been made in optimizing spectral and spatial feature extraction. Chhapariya et al. [43] introduced the deep spectral–spatial residual attention network (DSSpRAN), which employs a residual attention mechanism to extract spectral and spatial features in parallel, mitigating gradient vanishing and training instability caused by high-dimensional characteristics. Their spectral residual attention network (SRAN) optimizes the importance ranking of spectral bands through adaptive weight assignment, while the spatial residual attention network (SpRAN) enhances local consistency by learning neighborhood pixel correlations, effectively reducing classification errors. However, DSSpRAN exhibits high sensitivity to hyperparameters, resulting in increased training difficulty and a heavy tuning burden. It is also susceptible to input perturbations and, according to experimental results, demonstrates strong dependence on data volume, requiring 10% of the dataset for training. Zhang et al. [44] designed the spectral–spatial differential convolutional network (S2DCN), which combines differential convolution with traditional convolution to emphasize the extraction of local details and gradient information, thereby diminishing the impact of over-smoothing in deep networks on classification performance and enhancing the model's discriminative power in high-dimensional data. However, S2DCN also involves some critical hyperparameters, with model performance highly dependent on precise tuning. Additionally, the network lacks mechanisms to mitigate gradient fluctuations or enhance robustness against perturbations.

Based on this, we present glance and gaze attention (GGA) with a sorted texture-aware network (ST-GGNet) incorporating a budding growth optimizer (BGO) with the following key contributions.

1) To overcome the challenge of modeling long-term dependencies, we introduce a dual-branch parallel GGA mechanism for enhanced efficiency and effectiveness in long-range dependency modeling. GGA employs a multidimensional attention mechanism and innovatively integrates a KAN layer into the attention framework for HSI classification. Specifically, Glance-attention (Glance) performs global self-attention through comprehensive scanning of the HSI's receptive field, while Gaze-attention (Gaze) focuses on local, detailed information. The synergy between these components balances global and local information, effectively modeling long-term dependencies in input features and ensuring the preservation and adequate expression of key information across different bands.
2) To enhance the utilization of HSI information and mitigate the model's dependence on the number of training samples, we propose the sorted texture-aware module (STM) feature extraction module. This module deeply integrates spatial and spectral information at multiple scales, enhances texture and detail feature extraction through center differential convolution (CDC), and constructs graph structures based on spectral similarity. By thoroughly mining and efficiently leveraging spectral information in HSI, STM increases spectral data utilization and reduces the model's reliance on extensive training data.

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3) To overcome the tendency of the model to converge to local optima, we present a meta-heuristic optimization algorithm named BGO, which draws inspiration from the principles of bud growth mechanisms. This module inherits the concepts of adaptive learning rates, momentum, and employs a budding growth mechanism with Gaussian perturbations to enhance the model's exploratory capabilities during parameter optimization. This approach improves the exploration of superior solutions, especially under limited sample conditions.
4) Extensive evaluations on four leading HSI classification datasets demonstrate the algorithm's robustness and generalization ability. Comparative studies against nine state-of-the-art (SOTA) methods show that our proposed approach achieves highly competitive outcomes.

II. Related Work

A. Attention Mechanism for HSI Classification

Attention mechanisms enhance model performance by focusing on critical regions, amplifying useful features while suppressing irrelevant ones. In remote sensing imagery, noise and uneven distribution of ground objects can impair model performance. Effective attention mechanisms mitigate these issues by highlighting key regions, thereby improving model robustness and generalization [62]. Esmaeil et al. [61] introduced the multiscale attention fusion module, which integrates attention maps from multiple channels or levels. By employing positional encoding to optimize spatial attention, the module effectively captures spatial relationships among targets through cascaded attention modules with varying receptive fields. However, the use of convolutional layers alone fails to capture long-range dependencies. ReSC-net, proposed by Fu et al. [63], leverages a channel-attention-enhanced residual module and a spatial-channel joint attention mechanism to improve spatial feature representation in HSI, addressing the "curse of dimensionality" and mitigating overfitting. However, ReSC-net focuses more on importance compression than leveraging spectral structures, lacking explicit modeling of spectral sequences. The self-attention mechanism in Transformers is well-suited for processing long sequence data, aligning with the modeling requirements of the spectral dimension in HSI, which often comprises hundreds of bands [64]. Consequently, Transformers have found extensive use in HSI classification. Li et al. [1] developed the self-pooling transformer (SPFormer) architecture using channel shuffle and multihead self-pooling (CSSM-MHSP) and central token mixer (CTM) to replace the spectral and spatial dimensions in conventional Transformers. Wu et al. [81] introduced a central target token to adaptively learn semantic tokens from feature maps, enhancing focus on target pixels while preserving secondary yet discriminative spectral information. This design mitigates information loss caused by insensitivity to subtle variations. Liu et al. [82] reduced computational complexity by decomposing the spatial attention matrix in the Transformer architecture and further enhanced detail recovery by incorporating spatial–spectral priors-guided self-attention with attention masks derived from spatial and spectral cues. CSSM-MHSP restructures spectral features through self-outer product, channel shuffling, and multihead division, while CTM aggregates spatial information based on similarity between central pixels and their neighborhoods. However, CTM primarily focuses on central-neighborhood relations, neglecting intraneighborhood dependencies, and exhibits high computational complexity. To reduce the computational cost of global self-attention, Alkhatib and Jamali [65] proposed the local window attention (LWA) mechanism. LWA uses local attention instead of global attention, gradually expanding the receptive field within a hierarchical structure. However, LWA is influenced by CNN priorities and lacks inter-window interaction, potentially leading to information fragmentation. Xu et al. [55] utilized independent Vision Transformer (ViT) modules to extract spatial and spectral information separately, followed by the cross spatial–spectral attention (CS2A) module to fuse these features efficiently, capturing interdependencies between spatial and spectral features. Despite this, Transformer-based architectures tend to focus on global information, lacking effective strategies for local feature handling and posing high computational demands. To enhance spatial feature representation in high-dimensional data and alleviate the dimensionality curse, Zhao et al. [66] proposed the cosine and Gaussian–Euclidean similarity self-attention (CG-SA) mechanism. CG-SA effectively models the relationships between central and surrounding pixels, enhancing global spatial feature expression and addressing the dimensionality issue inherent in Euclidean similarity. However, cosine similarity is insensitive to the magnitude of feature vectors, neglecting the impact of feature size variations on distinguishing different classes.

Existing attention mechanisms predominantly focus on spatial feature extraction via CNNs or spectral sequence modeling via Transformers, either failing to capture global information or lacking efficient interaction mechanisms. This study proposes an attention mechanism that balances global and local feature interactions, ensuring critical spectral information is captured while suppressing irrelevant data, thus facilitating comprehensive modeling of long-term dependencies.

B. Feature Extraction of Spectral and Texture

Remote sensing imagery typically contains large volumes of raw data with high dimensionality, making direct use for classification or recognition inefficient. Extracting key features such as spectral [67], shape [68], spatial [69], and texture [79] can significantly reduce redundant information, preserving only the most relevant parts [70], thereby enhancing processing efficiency and the model's discriminative ability [71]. With the advancement of CNNs, spatial and shape feature extraction has reached a higher level of maturity [72]. To improve the efficiency of classification and recognition of remote sensing imagery, researchers have proposed various methods and strategies for spectral and texture feature extraction, as discussed below. Huang et al. [83] incorporated a superpixel-based graph into self-attention, embedding local spatial priors to reinforce spectral–spatial feature extraction under local spatial constraints.

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Meng et al. [73] proposed a multiscale feature extraction module based on traditional CNN, integrating convolutional kernels of three different scales. The module employs an Squeeze-and-Excitation (SE) mechanism to allocate channel weights, effectively retaining features with spectral differences and enhancing their representation. However, the framework primarily focuses on global context modeling, potentially lacking in detailed texture extraction. Ma et al. [74] developed a feature extraction method for high spatial resolution remote sensing images by combining an encoder-decoder architecture with the pixel dichotomy model (PDM) and kernel-driven model (KDM). This approach characterizes texture features through multiangle reflectance, though it is sensitive to parameter fitting strategies. Yang et al. [75] introduced the ITCNet model, which extracts multiscale features in a parallel manner within an image-level framework, integrating global spectral features via Transformer and local texture features via a CNN. Although the model effectively combines these features, it lacks scalability for large-scale images or multiclass scenarios. Guo and Liu [84] designed a lightweight multihead efficient attention in the Transformer-CNN architecture, using an agent token to reduce self-attention complexity and improve spectral sequence modeling. They further integrated GRU to enhance spectral representation and better capture intrinsic spectral dependencies. Zhang et al. [85] developed a tensor-based self-attention mechanism that directly operates on 3-D HSI tensors, eliminating the need for sequence flattening and preserving spectral structures while enhancing spatial–spectral interaction. Wang et al. [76] employed linear projection to extract query (Q), key (K), and value (V) tensors, partitioning continuous spectral bands into spectral segments through a channel partitioning mechanism. The dynamic routing method selects the most relevant spectral regions, and token-to-token spectral attention captures complementary information among bands. However, in HSI with significant noise, focusing excessively on fine-grained features may amplify noise. Ding et al. [9] proposed the local-global perception transducer (LGPT), combining multihead attention with depthwise separable convolution to differentiate noise from valid information. The multihead spectral–spatial depth attention (MH2SD) module addresses traditional transformers' limitations in capturing local textures, allowing for better perception of complex spectral and spatial features across bands. Nonetheless, LGPT's feature fusion primarily relies on concatenation or pointwise convolution, limiting interactive capacity. Liu et al. [78] proposed multilevel prototype alignment (MLPA), integrating prototype alignment modules at shallow, intermediate, and deep levels to achieve hierarchical feature alignment. This fine-grained alignment reduces the inconsistency between shallow (generic) and deep (task-specific) features. However, the lack of a modeling mechanism to distinguish alignments at different levels may result in ineffective or redundant feature alignment, especially when processing high-dimensional data or when object boundaries are connected or overlapping. Our proposed STM leverages multiscale convolution to capture target features at varying scales, effectively avoiding ineffective or redundant alignment. It models graph structures based on spectral similarity, enabling precise discrimination of distinct targets along connected edges.

III. Preliminary

In our model, KAN's nonlinear data transformation plays a crucial role in efficiently establishing long-term dependencies in GGA, while CDC is key to enabling the STM module to extract finer texture details. To better present our technical details, we will discuss them separately in this section.

A. Kolmogorov–Arnold Networks

Liu et al. [35] proposed KAN, which is a neural network model derived from the Kolmogorov–Arnold representation theorem. Unlike conventional MLPs, KAN employs learnable activation functions on the edges, replacing the fixed linear weights used in traditional models, thus achieving superior nonlinear transformations. This structure allows KAN to outperform traditional MLPs in function fitting tasks with fewer parameters, offering improved accuracy and interpretability. Inspired by these features, we argue that KAN's nonlinear data processing and learnable linear weights make it more suitable for the high-dimensional, limited sample tasks of HSI classification under few-shot conditions.

The Kolmogorov–Arnold theorem is a significant mathematical result on function approximation, stating that any continuous multivariable function can be approximated through a specific form of representation. As shown in Fig. 2(b2). Specifically, KAN replaces each linear weight with a 1-D spline function, enhancing both model accuracy and interpretability. Initially, the input feature X is normalized to X', scaling the features to the [0, 1] range

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min} + \epsilon} \tag{1}$$

where $X \in R^{N \times F_{\text{in}}}$ represents the input feature matrix, N is the number of samples, F_{in} is the number of input features, $X_{\max}$ and $X_{\min}$ denote the maximum and minimum values of each feature within the batch, and ϵ is a small constant to prevent division by zero.

B-splines represent complex curves or surfaces using control points and basic functions. Due to their locality and continuity, they are commonly used to efficiently approximate complex multidimensional data. By combining B-spline basis functions, the model can more accurately fit intricate data patterns. Therefore, B-splines of a specified order are computed using spline-based functions. For each normalized feature X', the spline basis function $S(X)$ is evaluated at predetermined knots k

$$S_{i,j}\left(X_i'\right) = \max\left(0, 1 - \left[X_i' - k_j\right]\right) \tag{2}$$

where k_j is the knot indexed by j, uniformly distributed within the interval [0, 1], and $S_{i,j}(X_i')$ represents the value of the spline basis function for the feature indexed by i at knot j.

Subsequently, complex linear and nonlinear transformations are applied to X to enhance feature expressiveness, as illustrated in the following equation:

$$Y = \sigma(X W_{\text{base}}) + \sum_{i=1}^{F_{\text{in}}} S\left(X_i'\right) W_{\text{spline},i} \tag{3}$$

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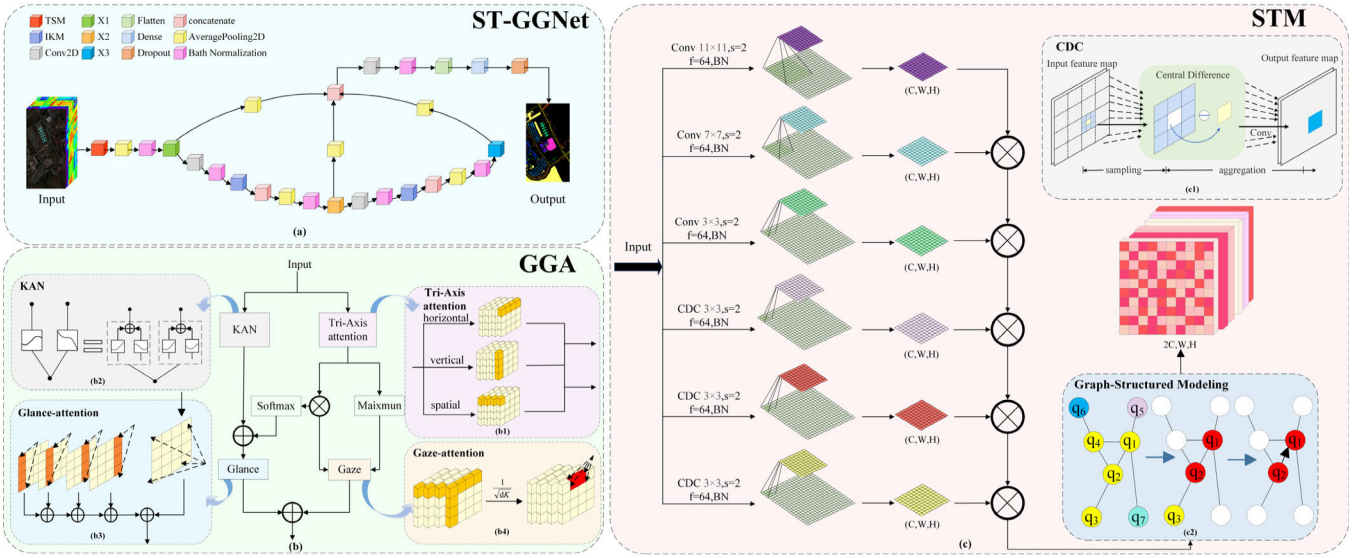


Fig. 2. Illustration of our proposed ST-GGNet. (a) Overall framework of ST-GGNet. (b) Overall framework of GGA. (b1) Depicts the structure of tri-axis attention, capturing input features along horizontal, vertical, and spatial dimensions. (b2) Shows the structure of the KAN layer transformation, where each node combines a base transformation with SiLU activation and a B-spline transformation. (b3) Illustrates the concept of Glance-attention, where global features obtained from the KAN transformation are combined with row-wise features to achieve effective large-scale modeling. (b4) Demonstrates the concept of Gaze-attention, focusing on local detailed features by magnifying features along different dimensions. (c) Overall framework of STM, which employs parallel multiscale convolutions followed by graph-based modeling. (c1) Demonstrates that CDC enhances edge sensitivity via convolutional kernel differencing on feature maps. (c2) Illustrates graph-structured modeling based on spectral similarity ranking.

$$\mathrm{SiLU}(x) = \frac{x}{1 + e^{-x}} \tag{4}$$

where $W_{\text{base}} \in R^{F_{\text{in}} \times F_{\text{out}}}$ is a trainable base weight matrix, F_{in} refers to the number of input features, and F_{out} refers to the number of output features, and X_i' denotes the feature column indexed by i of X', $W_{\text{spline},i} \in R^{K \times F_{\text{out}}}$ is the spline weight matrix corresponding to the feature indexed by i, K represents the number of knots in the spline. $S(X) \in R^{N \times F_{\text{in}} \times K}$ is the spline basis function matrix evaluated on the input features. σ is the activation function. To preserve nonlinear features and ensure smooth gradient propagation, we employ the SiLU activation function, as shown in (4).

Finally, to prevent overfitting in the high-dimensional space, we incorporate $L2$ regularization and entropy regularization into KAN. The goal of $L2$ regularization is to minimize the squared sum of weights, preventing excessively large weights that could lead to overfitting, as shown in (5). Entropy regularization, on the other hand, encourages a more uniform distribution of weights, thereby enhancing the model's robustness, as shown in (6)

$$\mathcal{L}_{L2} = \lambda_1 \left(\|W_{\text{base}}\|_2^2 + \left\|W_{\text{spline}}\right\|_2^2 \right) \tag{5}$$

where λ_1 is the weight decay coefficient for L2 regularization, while $\|W_{\text{base}}\|_2^2$ and $\|W_{\text{spline}}\|_2^2$ represent the $L2$ norms of the base weights and spline weights, respectively. This ensures that large weights are penalized through $L2$ regularization, thereby controlling the model's complexity

The goal of entropy regularization is to encourage a more uniform weight distribution, which enhances the model's robustness

$$\mathcal{L}_{\text{entropy}} = \lambda_2 \sum_i (-p_i \log(p_i)) \tag{6}$$

where p_i denotes the normalized value of the spline weights, and λ_2 is the weight coefficient for entropy regularization. Entropy regularization prevents excessive concentration or bias of the spline basis function weights in specific regions, reducing the model's bias.

The order of B-spline functions significantly influences model expressiveness and stability. Lower-order splines reduce complexity and overfitting risk but lack the capacity to model complex nonlinearities, causing unstable attention weights. Higher-order splines enhance flexibility but may overfit. Guided by experimental findings, cubic B-splines are employed in the KAN layer to balance expressive power and smoothness in parameter transitions.

B. Center Differential Convolution

Yu et al. [47] proposed CDC in the context of face anti-spoofing to enhance CNN's ability to represent fine-grained details, edge information, and textures. By focusing on the center difference, i.e., the gradient of pixel values in a local region, CDC effectively captures subtle features, such as the grid artifacts commonly seen in forged facial images, making it highly effective in face antispoofing. Zhan et al. [77] applied CDC to remote sensing detection to highlight the response of important features, which brought us inspiration in remote sensing image processing. Considering the multiple spectral bands in HSI, where each band represents a different wavelength, CDC not only efficiently extracts subtle spectral differences but also enhances the extraction of texture features in spatial information. The structure of the CDC is shown in Fig. 2(c1), and its mathematical formulation is provided in the following equation:

$$y_N(l_0) = \sum_{l_n \in R} w(l_n) \cdot x(l_0 + l_n) \tag{7}$$

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$$y_C(l_0) = \sum_{l_n \in R} w(l_n) \cdot x(l_0 + l_n) + \theta \cdot \left(-x(l_0) \cdot \sum_{l_n \in R} w(l_n) \right). \tag{8}$$

In these equations, R represents the local receptive field in the input feature map X, and y signifies the output feature map. l_0 indicates the relevant positions on the input and output feature maps, while l_n traverses the positions within R. w represents the convolution kernel weights. The hyperparameter θ controls the influence of the central position l_0 on the overall output. By adjusting θ, the weight of the central pixel relative to surrounding pixels can be modulated. As θ increases, the centrality of the CDC effect is enhanced, and when θ is zero, it is equivalent to standard convolution, as shown in (7). Through comprehensive experiments, we determined that $\theta = 0.2$ is the optimal hyperparameter value.

Compared to using only CNNs for feature extraction, CDC more effectively captures subtle and essential structural features in HSI, such as spectral details and texture variations. This method emphasizes edge and structural details, highlighting differential information in fine features, which improves adaptability in complex environments and reduces misclassification at land cover boundaries due to feature similarity. It also enhances feature representation and the use of spectral information.

IV. ST-GGNet

This chapter begins by outlining the operational workflow and structural framework of ST-GGNet. It then provides a detailed explanation of the GGA network, which leverages KAN layers and feature interactions for effective long-range dependency modeling. Next, it introduces the STM feature extraction module, which utilizes multiscale differential convolutions to capture textures and graph-structure modeling sorted by spectral similarity. Finally, the chapter explores the BGO, a metaheuristic algorithm inspired by the natural growth processes of buds, designed to enhance optimization efficiency.

A. ST-GGNet

To improve the model's capacity to capture complex long-range dependencies, fully exploit land feature information from the rich data in HSI, reduce reliance on training sample size, and ensure stable convergence to optimal solutions, we propose the ST-GGNet to enhance HSI classification performance. Fig. 2 illustrates the overall architecture of the ST-GGNet.

It comprises three primary modules: GGA, STM, and BGO. The GGA module performs nonlinear transformations on input feature maps using KAN and employs Tri-Axis Attention for feature cross-fusion. It then leverages the collaboration of Glance-attention (Glance) and Gaze-attention (Gaze) to attention to facilitate effective long-term modeling, preserving key information across distant spectral bands while enhancing the representation of useful features and suppressing redundant information. The STM module fully utilizes multiscale feature information in HSI by applying CDC to better capture details and edges, enabling more thorough feature extraction of land objects. It subsequently models the graph structure using a neighborhood matrix based on spectral similarity to improve the utilization of spectral information. Considering the high-dimensional complexity of HSI data, which can lead to local optima during training, the BGO module introduces a budding growth mechanism to stabilize convergence while effectively exploring superior solutions. The workflow of ST-GGNet is illustrated in Fig. 2.

B. GGA Mechanism

To address the challenge of effectively modeling long-term dependencies in HSI, this article introduces a self-attention mechanism called GGA. The process begins with Tri-Axis Attention, which extracts multichannel information from input features across horizontal, vertical, and spatial dimensions. This step enhances the model's ability to capture features from diverse perspectives. Subsequently, Glance is applied to assimilate information at a global level, facilitating the modeling of extensive dependencies and capturing key cross-spectral features. Although Glance efficiently captures long-range feature representations, it may overlook relationships and interactions within certain local regions. To address this, Gaze is introduced, focusing on critical local information to capture the interactions between HSI bands and pixels that Glance might miss. By integrating Glance and Gaze in parallel, GGA successfully balances global and local feature learning. This collaborative approach preserves essential details while avoiding the loss of critical information and the inclusion of irrelevant data, enabling efficient and accurate long-range dependency modeling. The structure of GGA is shown in Fig. 2(b).

1) Tri-Axis Attention: HSI encompasses a wide array of spectral channels, resulting in significant redundant information. To enhance the neural network's ability to assimilate the multidimensional features of spectral images, we introduce Tri-Axis Attention, which integrates horizontal, vertical, and spatial attention mechanisms, as illustrated in Fig. 2(b1). This design enables the model to proficiently identify and meticulously process the most pertinent features within the input data. The precise procedure is described by (9)–(11), where the variables Q, K, and V represent the horizontal, vertical, and spatial feature information, respectively,

$$Q_{i,j,k} = \frac{X_{i,j,k}}{\sum_{i'=1}^{p_1} \exp\left(W \otimes X_{i',j,k}\right)} \tag{9}$$

$$K_{i,j,k} = \frac{X_{i,j,k}}{\sum_{j'=1}^{p_2} \exp\left(W \otimes X_{i,j',k}\right)} \tag{10}$$

$$V_{i,j,k} = \frac{X_{i,j,k}}{\sum_{k'=1}^{p_3} \exp\left(W \otimes X_{i,j,k'}\right)} \tag{11}$$

where $\otimes$ represents the dot product operation, and W denotes the weight matrix for each Softmax layer. The exponents i, j, k correspond to the width, height, and depth dimensions along the horizontal, vertical, and spatial axes, with p_1, p_2, and p_3 denoting their respective extents. The first step involves decomposing the image along these three axes, generating feature maps that undergo element-wise multiplication with the weight matrices. The exponents of these products are used as

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normalization factors. Through the summation normalization process, normalized scores are computed for each component of Q, K, and V. These scores represent the contribution of each input element to the output, helping the model identify the most significant features. This process highlights relevant features while minimizing redundancy, enabling more efficient feature representation and improving HSI classification accuracy. By applying the softmax function along different axes of the input tensor, the model isolates important features in each dimension (horizontal, vertical, or spectral spatial) and leverages these features to develop more sophisticated hierarchical data representations for classification purposes.

2) Glance-Attention: Traditional MLP architectures employ fixed activation functions at neurons, which hampers their ability to uncover complex nonlinear relationships within global information when processing high-dimensional HSI feature maps. Consequently, effective long-range dependency modeling becomes challenging. Compared with MLPs that derive nonlinearity solely from fixed activation functions, KAN offers a higher degree of weight learnability. Inspired by this approach, we apply a KAN layer to nonlinearly process the input feature map X, thereby effectively capturing global information in HSI feature maps, as shown in Fig. 2(b2). This enables the attention mechanism to engage in deeper interactions and fusion within the feature space, improving the model's comprehension of HSI data and, consequently, classification accuracy. The operational mechanism and theoretical basis of the KAN layer are detailed in Section III-A.

As illustrated in Fig. 2(b3), the Glance-attention branch receives input feature maps processed by the KAN layer, as detailed in (12) and (13). Global denotes the spectral global information, where Glance refers to the scanning information of HSI

$$\text{Global} = \frac{\exp\left(Q_{i,j,k} \otimes K_{i,j,k}\right)}{\sum_{k'=1}^{p_4} \exp\left(Q_{i,j,k'}\right)} \tag{12}$$

$$\text{Glance} = \text{concatenate}[Y, \text{Global}] \tag{13}$$

where p_4 represents the number of spatial channels. Y represents the feature map transformed by KAN. To minimize parameter count and mitigate noise in the amplitude domain, the Glance-attention branch initially applies a Softmax function to normalize feature information across the horizontal, vertical, and spatial dimensions within tri-axis attention. This normalization facilitates comprehensive image scanning and precise acquisition of global information. Subsequently, the original input is combined with self-attention weighted features, maintaining spatial correlation integrity while preserving global contextual essence. The Glance-attention mechanism is designed to capture global information, allowing for comprehensive long-range representations of HSI. This enables efficient self-attention within a wide receptive field, enhancing the model's ability to identify long-distance dependencies.

3) Gaze-Attention: Although the Glance branch effectively captures long-range representations, it overlooks cross-region local connections and neglects valuable local information essential for vision tasks dependent on local cues. In processing high-dimensional spatial data of HSI, local information encompasses rich spatial and texture features surrounding each pixel. To compensate for the reduced local detail information between hyperspectral bands and pixels, we designed the Gaz mechanism, as shown in Fig. 2(b4). As detailed in (14) and (15), Local refers to spectral local details, while Gaze denotes the focused information within HSI

$$\text{Local}_{i,j,k} = \max\left(Q_{i,j,k}, K_{i,j,k}, V_{i,j,k}\right) \tag{14}$$

$$\text{Gaze} = \frac{Q_{i,j,k} \otimes K_{i,j,k} \otimes V_{i,j,k}}{\exp\left(-\frac{\text{Local}_{i,j,k}}{\sqrt{dk}}\right)}. \tag{15}$$

As shown in (15), a scaling factor is proposed to enable the model to perform scaled dot-product attention. This ensures selective interaction with information aspects essential to the primary task. By establishing a scaling ratio of $(1/\sqrt{dk})$, and normalizing the feature maps of Q, K, and V vectors through exponential operations, the focus intensity of features is calibrated, where dk represents the dimension of Q, K, and V vectors. This precaution prevents any single attribute from dominating due to scalar differences, thereby ensuring focused attention on relevant regions. This scaling mechanism is a crucial component of the Gaze architecture, playing a vital role in computing dot product among Q, K, and V, especially in high-dimensional contexts. Without this adjustment, the Softmax gradient could diminish, hindering effective learning. To address this, max operation is applied to the maximum values of the Q, K, and V vectors, which enhances the vectors with stronger correlations and increases the model's sensitivity to spatial and texture features. This process, as shown in (14), in detail in different dimensions of features processed by tri-axis attention, is amplified and then scaled according to (15). This maintains attention scores within an optimal range, avoids excessively large values in dot-products, and facilitates efficient scaling of attention metrics. The Gaze branch focuses on the most informative regions by applying dimension-based scaling factors of feature space, compensating for any loss of detail during the Glance's global scanning process

$$\text{GGA} = \text{concatenate}\left[\alpha^{*}\text{Glance}, \beta^{*}\text{Gaze}\right]. \tag{16}$$

Finally, we employ the method in (16) to assign weights to Glance and Gaze. Through the synergistic effect of these two mechanisms, GGA effectively integrates feature information across different ranges, enabling long-range dependency modeling based on feature interactions. This approach adaptively enhances significant spectral and spatial features while preserving necessary local details, thereby preventing the loss of critical information and fully expressing key features to improve classification performance. Particularly in edge information classification, it effectively retains and focuses on essential features. Experimental results validate its capability to significantly enhance model robustness.

C. Sorted Texture-Aware Module

In HSI, significant differences among objects are primarily evident in their texture and structural characteristics. Consequently, effectively capturing relevant information with limited data poses a critical and challenging task. When employing CNNs for classification, images are typically partitioned into local regions through convolution operations to extract spatial

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features from these blocks [34]. However, HSI classification involves classifying each pixel 25, leading CNN-based feature extraction to rely on fixed-size convolution kernels, which restrict the region sizes. To capture multiscale object features in HSI, we proposed the phantom fractal module (PFM) in [46], utilizing convolution kernels of varying sizes to extract features at different scales. This approach enhances classification performance compared to fixed convolution sizes. Nevertheless, it inadequately leverages spectral differences to distinguish spatially adjacent similar features, resulting in misclassification of neighboring objects. To resolve the problem, we developed STM as the feature extraction module of the network, building upon the foundation of PFM, as illustrated in Fig. 2(c).

In this work, we use multiscale convolutions with CDC to build a neighborhood matrix based on spectral information correlations for graph-based structural modeling. This method establishes spatial long-range dependencies through spectral data, helping to avoid misclassification due to spatial feature similarities and improving the use of spectral features. Initially, multiscale parallel convolutions are applied to the input image to extract feature information at various scales, as shown in the following equation:

$$Y_{\mathrm{STM}} = y_N\left[\mathrm{concatenate}\left[y_{N1}, y_{N2}, y_{N3}, y_{C4}, y_{C5}, y_{C6}\right]\right] \quad (17)$$

where $y_{N1,2,3}$ denotes the standard convolution layers at the first to third scales, while $y_{C4,5,6}$ corresponds to the CDC layers from the fourth to sixth layers. The operational mechanism and theoretical basis of the CDC are presented in Section III-B. In contrast to the traditional convolution articulated, the CDC integrates both intensity and gradient information, significantly excelling in the articulation of granular local spatial details. The integration of the CDC operator facilitates the genesis of the STM. By combining multiscale standard convolutions with CDCs in a parallel configuration, the network processes feature information at various scales, enriches feature representation, and integrates both intensity and gradient information.

HSI contains reflectance information across multiple spectral bands for each pixel, and adjacent land covers in an image often share similar spatial features, such as edge shapes. However, neighboring pixels contain high-dimensional spectral information that effectively distinguishes different land cover characteristics. To fully harness this spectral information across multiple scales, we innovatively integrate multiscale convolutions with graph structure modeling. Using the feature maps generated by parallel multiscale convolutions, we construct a neighborhood matrix based on spectral similarity and input it into GCNs. By sorting features, we differentiate land covers with distinct spectral characteristics, thereby enhancing the discriminative ability between adjacent land covers. Specifically, the fused features are first reshaped into a 2-D node set, where each node corresponds to a pixel location and its feature vector in the feature map, as shown in the following equations:

$$\boldsymbol{x}_i = \boldsymbol{F}_{h_i,w_i}, \quad i = 1, 2, \ldots, N \quad (18)$$

$$\boldsymbol{X} = \mathrm{reshape}(\boldsymbol{F}, (N, C)) \quad (19)$$

where $F \in R^{H\times W\times C}$ denote the fused feature map, which is reshaped into a node feature matrix $X \in R^{N\times C}$, H and W represent the height and width of the feature map, respectively, C is the number of feature channels, and $N = H \times W$ denotes the total number of nodes. Each vector x_i corresponds to a pixel location (h_i, w_i) in the feature map. For any two nodes i and j, a neighborhood matrix is constructed based on the spectral similarity between their feature vectors, as described in the following equation:

$$\begin{cases} d_{i,j} = x_i - x_{j2} \\ N(i) = \left\{ j | j \neq i, \text{The } k \text{ node with the smallest } d_{i,j} \right\} \\ A_{i,j} \begin{cases} 1 \\ 0. \end{cases} \end{cases} \quad (20)$$

The spectral distance between computing nodes i and j is denoted as d. For each node i, the k closest nodes based on spectral distance are chosen as its neighbors, and their indices are denoted by $N(i)$. Using these neighbors, the adjacency matrix $A \in \{0, 1\}^{N\times N}$ is constructed, where $A_{i,j} = 1$ indicates that node j is a neighbor of node i due to their similarity. Thus, matrix $A_{i,j}$ captures the linkage structure among nodes in the graph.

In contrast to conventional local convolution, graph structure modeling allows similar spectral features of spatially distant pixels to interact within the feature space, capturing long-range spectral correlations and enhancing spectral utilization. Graph convolution relies on the adjacency matrix to define relationships between nodes, enabling feature aggregation and thereby enriching feature representations, as illustrated in the following equation:

$$H^{(l+1)} = \sigma\left(AH^{(l)}W^{(l)}\right) \quad (21)$$

$$elu = \begin{cases} O_{i,j}, & O_{i,j} > 0 \\ \alpha\left(e^{O_{i,j}} - 1\right), & O_{i,j} \leq 0 \end{cases} \quad (22)$$

where $H^{(l)}$ denotes the feature matrix at layer l, $W^{(l)}$ is the learnable weight matrix, σ represents the activation function, and O signifies the graph-related output matrix. The adjacency matrix is constructed based on spectral similarity, achieving sparse connections by selecting the most similar nodes. This approach facilitates more efficient feature propagation, captures spectral correlations between distant nodes, and enhances interactions among similar features. We utilize elu as the activation function for the graph convolution, as shown in (22), to retain negative information and perform nonlinear mapping. This not only diversifies feature expressions but also alleviates the vanishing gradient problem, thereby enhancing the stability of model training.

D. Budding Growth Optimization

In HSI classification tasks, the complexity of spectral information and the diversity of spatial features play a crucial role in determining the performance of classification networks. Choosing the right optimizers, such as stochastic gradient descent (SGD) and Adam [48], can improve model convergence speed and classification accuracy by optimizing

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the parameter update process. SGD accelerates learning by using stochastic sampling and momentum, which reduces oscillations and facilitates the search for optimal solutions in high-dimensional data spaces. On the other hand, Adam employs first and second-order exponential moving averages of gradients to perform precise learning rate adjustments and normalizes using the EMA of gradient squares for bias correction, thereby improving the adaptability of parameter updates. However, the high-dimensional nature of HSI data often leads to significant class overlap, causing gradient descent-based optimization algorithms to become trapped in a local minimum. Furthermore, nonlinear features and noise in high-spectral data may destabilize gradient signals, impairing adaptive optimizers' ability to accurately estimate gradient directions, which hinders model learning and generalization.

Therefore, the rational selection and tuning of optimizers and their parameters are crucial for HSI classification tasks. In contrast, RMSProp, an adaptive learning rate optimization algorithm, dynamically adjusts learning rates by maintaining a moving average of squared gradients for each parameter, accommodating the gradient distribution and update requirements of different bands or features. However, RMSProp lacks an effective exploration mechanism and relies heavily on historical gradient information, increasing the risk of becoming trapped in local minima, especially in highly nonconvex loss landscapes with steep or flat saddle points, thereby significantly affecting final classification performance.

Inspired by [49] and the typical growth behavior of young plant stems, which grow toward light but slightly sway due to environmental disturbances to adjust their growth direction, we propose a BGO based on root mean square propagation. BGO introduces an exploration mechanism to the traditional RMSProp optimizer, enhancing its ability to escape local minima during model training. In this approach, gradients are treated as indicators guiding model parameters toward optimal solutions, while the algorithm simulates the random extension and directional selection of bud growth by dynamically adjusting the parameter update path with Gaussian perturbations. This mirrors the plant's adjustment to light. By enabling a broader search in high-dimensional, complex parameter spaces, the optimizer can explore new directions during updates, effectively reducing the risk of getting stuck in local optima and promoting the pursuit of global optimal solutions.

Following the initialization of the optimizer's hyperparameters, BGO assigns individual slots to each model parameter. It determines the gradient g_t by differentiating the target loss function $J(\theta_t)$ with respect to θ_t as (23). The rms update mechanism is presented in (24). To dynamically modulate the learning rate and minimize update steps for parameters experiencing significant gradient g_t variations, BGO utilizes the learning decay rate ρ to regulate the impact of the current gradient g_t on the rms accumulator R_t, which tracks the exponential moving average of gradient squares.

Building upon this foundation, we introduce a bud growth mechanism as delineated in (26) and (27). For the current growth factor c_t, external disturbances to the bud's growth direction are simulated using a perturbation factor f_d. Specifically, f_d modulates the influence of the preceding growth factor c_{t-1} on c_t through the growth factor decay rate γ, thereby perturbing the current gradient. The perturbation coefficient α governs the magnitude of f_d's impact on the growth factor. Additionally, d represents a Gaussian perturbation term as defined in (25), where μ denotes the mean offset of the perturbation and σ determines its dispersion, the perturbation amplitude. In summary, the current growth factor c_t is jointly influenced by f_d and the previous growth factor c_{t-1}, with their respective effects controlled by γ and α. Subsequently, the bud's growth state is updated according to (28) by adding c_t to the prior growth state s_{t-1}, yielding the current growth state s_t. This velocity facilitates the smooth updating of parameter values during bud growth. The growth coefficient λ regulates the contribution of s_{t-1} to s_t. Due to the stochastic influence of f_d on the growth factor, the growth state update inherently incorporates randomness. This stochastic parameter update mechanism effectively enhances the model's exploration capabilities, enabling it to escape local minima and thereby improving the global search performance of the optimization process

$$g_t = \nabla_{\theta_t} J(\theta_t) \tag{23}$$

$$R_t = \rho \cdot R_{t-1} + (1-\rho) \cdot g_t^2 \tag{24}$$

$$d \sim N\left(\mu, \sigma^2\right) \tag{25}$$

$$f_d = d \cdot g_t \tag{26}$$

$$c_t = \gamma \cdot c_{t-1} + \alpha \cdot f_d \tag{27}$$

$$s_t = \lambda \cdot s_{t-1} + c_t. \tag{28}$$

Finally, the comprehensive parameter update rule is presented in the following equation:

$$\theta_t = \theta_{t-1} - \frac{\eta}{\sqrt{R_t + \epsilon}} \cdot (g_t) - \frac{\eta}{\sqrt{R_t + \epsilon}} \cdot (s_t) \tag{29}$$

where ϵ is a small constant to prevent division by zero. The gradient g_t is scaled by the rms accumulated value R_t, adaptively adjusting each parameter's learning rate η. Specifically, the learning rate decreases in regions with larger gradients to enhance learning and increases in areas with smaller gradients to accelerate convergence. The core mechanism of the algorithm introduces a bud growth process with Gaussian perturbation factors, utilizing the growth state term s_t to explore and expedite gradient descent. The detailed implementation process is outlined in Algorithm 1.

The traditional RMSProp optimizer lacks a corresponding mechanism to facilitate parameter space exploration. Often resulting in the algorithm becoming trapped in local optima, which limits generalization. To address this, we introduce a growth rule inspired by the dynamic adjustment of plant shoots, which sway due to random disturbances. By incorporating Gaussian noise into the parameter updates, we enhance the algorithm's ability to explore more effectively and efficiently. This approach not only expands the solution space, preventing premature convergence to local optima, but also improves performance in complex environments with land covers that have fuzzy boundaries. Additionally, it boosts the model's generalization ability, allowing it to better adapt to varying ground objects.

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Algorithm 1 Pseudocode of BGO

Input:
1: Cross-entropy function for loss calculation
3: learning decay rate ρ and learning rate η
2: List of model trainable parameters: $[p_0, p_1, \ldots, p_i, \ldots, p_n]$
Output:
1: Final updated parameter values
Initialization:
1: Gaussian perturbation mean offset μ and dispersion σ
2: Factor decay rate α
3: growth factor decay rate γ
4: Growth coefficient λ
Training Phase:
Initialize the R, c and s matrix to a zero matrix
For param p_i **in** $[p_0, p_1, \ldots, p_i, \ldots, p_n]$:
1: Calculate the gradient g_t by (23)
2: Update rms accumulation R_t by (24)
3: Compute the perturbation factor f_d using Gaussian noise by (25) and (26)
4: Update the growth factor c_t by (27)
5: Update the growth state s_t by (28)
6: Calculate the parameter θ_i update by (29)
7: Append the updated R_i, c_i, s_i, and p_i' to their respective lists.
End for: Return the list of updated parameters.

V. Experiment

A. Experimental Environment and Settings

To eliminate variations in test outcomes caused by differing experimental settings and ensure program stability and reproducibility, all experiments in this study were conducted within a uniform software and hardware environment. The specific configurations are detailed in Table I. Considering training duration and hardware capabilities, all subsequent experiments utilized a batch size of 32, an initial learning rate of 0.001, a decay rate of 0.001, and 64 training epochs. We set the number of runs to 10, meaning each experiment was repeated ten times to gather the necessary data for this study. Parameter settings for comparison methods followed the recommendations from their original publications. All other parameter settings for the comparison methods adhered to the recommendations outlined in their original publications.

B. Datasets Description

1) Pavia University Dataset: To reflect the model's stability and reliability under complex backgrounds, we selected this dataset as one of the datasets for comparison experiments. This dataset is part of the 2003 HSI collection from Pavia, Italy, captured using the ROSIS sensor. Initially captured across 115 spectral bands, 12 noisy bands were excluded, resulting in 103 usable bands. It has a spatial resolution of 1.3 m and ground-truth (GT) data comprising nine classes. Notably, the dataset features complex background pixels, sparse connected regions, and numerous isolated pixels. Combined with its imbalanced class distribution, these characteristics make it a challenging dataset for evaluating the model's robustness and its ability to distinguish boundary regions, with classification results potentially being significantly influenced by random factors.

2) Salinas Dataset: To reflect the model's dependence on the data, we selected this dataset as one of the datasets for comparison experiments. This dataset, captured using the AVIRIS sensor in California, consists of 512 × 217 pixels with a spatial resolution of 3.7 m per pixel and includes 224 spectral bands. It features 16 distinct object classes. Compared to the other three HSI datasets, this dataset has a larger number of land cover classes but a relatively balanced distribution, making it suitable for testing the model's small-sample generalization ability when the sample size is significantly small (e.g., 0.5%).

3) Houston Dataset: To verify the model's overall performance and generalization ability when handling large-scale, multiclass, high-dimensional data, we selected this dataset as one of the datasets for comparison experiments. Collected in 2013 by the CASI-1500 hyperspectral imaging sensor at the University of Houston, Texas, this dataset covers the university and its surrounding areas. It has a spatial resolution of 2.5 m and spans spectral wavelengths from 380 to 1050 nm across 144 bands. The dataset includes 15 land cover classes. Due to its diverse categories and coverage, especially in terms of its performance in dealing with spatially continuous regions and the boundary ambiguity between different land covers, the Houston dataset is more complex and larger than the other three datasets. It is currently a widely recognized and challenging dataset.

4) WHU-Hi Dataset: To evaluate the model's classification performance under extremely low-sample scenarios, we selected this dataset as one of the datasets for comparison experiments. The WHU-Hi dataset, shared by the RSIDEA research group at Wuhan University, serves as a benchmark for precision crop classification and HSI studies. It includes three distinct UAV-based hyperspectral datasets: WHUHi-Longkou, WHUHi-Hanchuan, and WHUHi-Honghu. These datasets were collected using the headwall nano-hyperspace sensor mounted on a UAV platform over various agricultural regions in Hubei Province, China, featuring nanometer-level spectral and centimeter-level spatial resolution. Classifying the WHU-Hi dataset remains challenging due to the limited proportion of training samples. Specifically, the WHU-Hi-Longkou subset, collected in Longkou Town, Hubei Province, in 2018, includes a 550 × 400 pixel HSI with a spatial resolution of approximately 0.463 m, covering 270 bands from 400 to 1000 nm and categorized into nine classes. The scene encompasses an agricultural area cultivating six crop types: corn, cotton, sesame, broad-leaf soybean, narrow-leaf soybean, and rice. It has a significant class imbalance issue, making it more challenging to train with extremely limited samples (e.g., 0.2%) on this dataset compared to using the small-sample Salinas dataset.

Comprising both urban (Pavia University, Houston) and agricultural (Salinas, WHUHi-Longkou) scenes, the selected datasets differ markedly in spectral properties, spatial resolution, and land-cover types. This heterogeneity facilitates a thorough analysis of the model's robustness and generalization

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TABLE I
HARDWARE AND SOFTWARE PARAMETERS

Hardware environment		Software environment	
CPU	Intel(R) Xeon(R) Platinum 8255C CPU @ 2.50GHz	OS	Linux
RAM	40GB	CUDA	9.0
Video memory	11GB	Python	3.6.5
GPU	NVIDIA GeForce RTX 2080 Ti	Tensorflow	1.10.0
server platform	autodl	Keras	2.2.0

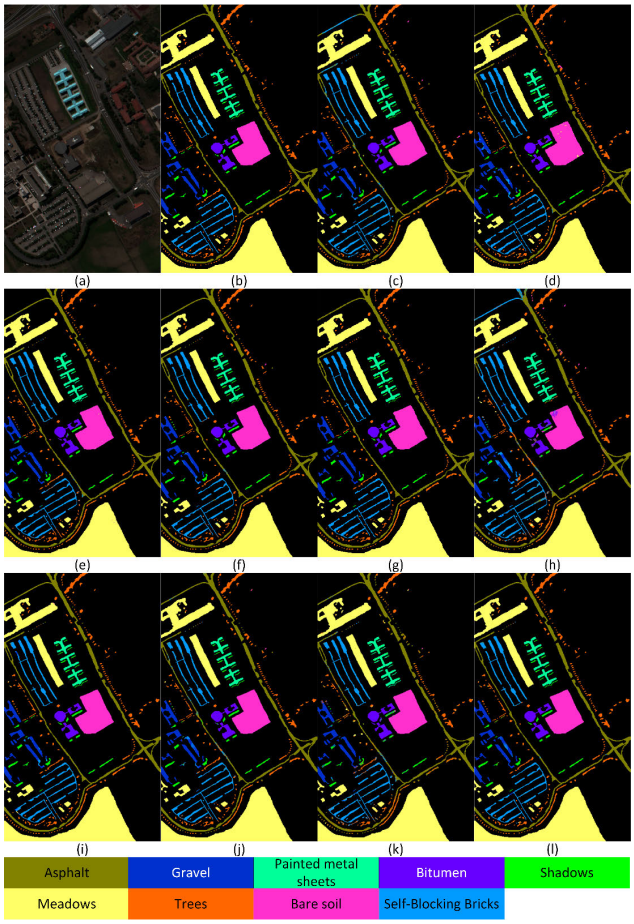


Fig. 3. Visualization results of comparative experiments on Pavia University. (a) Input. (b) GT. (c) SSRN. (d) DBDA. (e) HybridSN. (f) DGPF-RENet. (g) CS2DT. (h) Swin-Transformer. (i) GTFN. (j) GSC-CCTA. (k) SSFTT-KAN. (l) ST-GGNet.

across diverse scenarios. To thoroughly describe these datasets, Table II enumerates the categories and sample sizes. Due to layout constraints, some category names in Table II are abbreviated, but the complete names are available in Figs. 3–6.

To precisely illustrate the small-data dependency of ST-GGNet, only 0.2% of samples from the Salinas dataset were utilized as the training set in module effectiveness experiments and ablation experiments.

C. Evaluation Index

Overall accuracy (OA) is the proportion of correctly predicted samples relative to the total sample count. The calculation formula of OA is shown in the following equation:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FN} + \mathrm{FP} + \mathrm{TN}}. \tag{30}$$

In this context, TP indicates true positives where the model accurately predicts positive samples, TN indicates true negatives for correctly predicted negative samples, FP represents false positives where negative samples are misclassified as positive, and FN represents false negatives where positive samples are misclassified as negative.

Average accuracy (AA) reflects the mean classification accuracy across categories. It is determined as follows:

$$\mathrm{AA} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{recall}_i \tag{31}$$

where recall_i signifies the classification accuracy of category i, and n denotes the total number of categories.

Kappa, commonly denoted by k, is a comprehensive metric based on the confusion matrix

$$k = \frac{N\sum_{i=1}^{r} x_{ii} - \sum_{i=1}^{r}(x_{i+}{}^{*}x_{+i})}{N^2 - \sum_{i=1}^{r}(x_{i+}{}^{*}x_{+i})}. \tag{32}$$

In the formula, r represents the total number of classes; x_{ii} denotes the number of correctly classified instances; x_{i+} and x_{+i} signify the total number of the total pixels in the row i and column i, respectively; and N denotes the overall number of pixels employed for accuracy evaluation.

All subsequent experiments utilize AA, OA, and Kappa as evaluation metrics to assess model performance.

D. Comparison of ST-GGNet With SOTAs

To further evaluate performance, we compared our proposed method against nine SOTA models. The comparison includes CNN-based models: SSRN (2018) [50], HybridSN (2020) [51], and DGPF-RENet (2022) [46]; graph-structured models: GSC-CCTA (2023) [52] and GTFN (2023) [53]; attention-based models: DBDA (2020) [54], CS2DT (2023) [55], and Swin-Transformer (2023) [56]; and KAN-based model: SSFTT-KAN (2024) [38]. To evaluate the effectiveness of the proposed method, preliminary experiments were conducted

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TABLE II
DATASETS PARTITION FOR HSIS DATASETS

Pavia University		Salinas		Houston		WHUHi-Longkou	
Class	**Samples**	**Class**	**Samples**	**Class**	**Samples**	**Class**	**Samples**
Asphalt	6631	Br-we-1	2009	Health-g	1251	Corn	34511
Meadows	18649	Br-we-2	3726	St-gr	1254	Cotton	8374
Gravel	2099	Fallow	1976	Sy-gr	697	Sesame	3031
Trees	3064	Fa-pl	1394	Trees	1244	Br-le-so	63212
PM-sheets	1345	Fa-sm	2678	Soil	1242	Na-le-so	4151
Bare soil	5029	Stubble	3959	Water	325	Rice	11854
Bitumen	1330	Celery	3579	Res.	1268	Water	67056
SB-Bricks	3682	Gr-un	11271	Comm.	1244	Ro&ho	7124
Shadows	947	So-vi-de	6203	Road	1252	Mi-we	5229
		Co-se-we	3278	Hi-ways	1227		
		Le-4wk	1068	Ra-ways	1235		
		Le-5wk	1927	Park-Lot1	1233		
		Le-6wk	916	Park-Lot2	469		
		Le-7wk	1070	Ten-court	428		
		Vi-un	7268	Run-track	660		
		Vi-ve-tr	1807				

on four HSI datasets: Pavia University, Salinas, Houston, and WHUHi-Longkou, demonstrating the model's classification capabilities. To mitigate potential experimental errors, we recorded the mean values by repeating each experiment ten times, minimizing discrepancies and retaining two decimal places. For the classification result figures, we selected the outcomes with OA values closest to the final mean from the ten trials for presentation.

1) Results on the Pavia University: In this study, we randomly selected 5% and 10% of samples from each class as the training set for experimental analysis. Despite the limited labeled data, most models performed well, as shown in Table III. Additionally, Fig. 3 displays the classification results of ten models on the Pavia University dataset using a 5% training set. Analysis of Table III and Fig. 3 reveals that Transformer-based attention mechanisms, particularly the Swin-Transformer, performed slightly worse on this dataset. This underperformance could be due to Transformers' reliance on large datasets and their use of global self-attention mechanisms, which may not fully capture local features and could introduce additional noise. This hypothesis is supported by the presence of noise points in Asphalt, Bare Soil, and Self-Blocking Bricks within their visualization results. In contrast, our method achieved the highest classification accuracy among all compared approaches for these three land cover classes. This superior performance is likely attributed to our STM feature extraction module, which uses a neighborhood-based graph convolution structure. This design effectively distinguishes adjacent land cover types by analyzing spectral similarity variations during feature extraction. Consequently, our approach delivered SOTA performance across all three metrics, achieving an OA of 99.42%, AA of 98.90%, and Kappa coefficient of 99.24% with only 5% of training samples. Fig. 3 further demonstrates that ST-GGNet's results closely align with the GT map under conditions of limited training data. Moreover, while other models show noticeable noise in the bare soil, trees, and asphalt classes, ST-GGNet's visualization results are free from such noise, underscoring its effectiveness in redundancy suppression and feature representation.

2) Results on the Salinas: The Salinas dataset exhibits a more balanced sample distribution and presents lower classification difficulty. To enhance the classification challenge and thoroughly evaluate the models' dependence on limited training samples, we randomly selected 0.5% and 1% of samples from each class as training sets for experimental analysis. The test results are presented in Table IV. Notably, when utilizing 1% of the dataset, all compared models achieved high AA and OA. However, with only 0.5% of the training samples, CNN-based models SSRN and HybridSN, recorded OA values below 90%, specifically 88.32% and 89.20%, respectively. This indicates that deep models like SSRN are more sensitive to extremely small datasets, as their residual network structures require more samples to prevent overfitting. Additionally, HybridSN's strengths in spectral–spatial information fusion are diminished when dealing with highly similar land covers, as redundant information between adjacent spectral bands disrupts feature extraction.

Fig. 4 displays the classification results of ten models on the Salinas dataset using a 0.5% training set. It is evident that classifications for Grapes-untrained and Vineyard-untrained categories are confused due to their similar green vegetation spectral characteristics and overlapping key spectral bands. In contrast, graph-structured models GSCTA, GTFN, and our proposed ST-GGNet outperform others in distinguishing similar land covers, reaffirming the effectiveness of the graph-structured learning module in STM for classifying similar land cover types. As shown in Table IV, ST-GGNet achieves the highest scores in OA, AA, and Kappa metrics among all compared methods. This may be attributed to the adaptive feature enhancement provided by GGA, which effectively captures the unique characteristics of different land

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TABLE III
QUANTITATIVE RESULTS OF COMPARATIVE EXPERIMENTS ON PAVIA UNIVERSITY

Train Sample	Class	Model									
		SSRN	DBDA	Hybrid SN	DGPF-RENet	CS2DT	Swin-Transformer	GTFN	GSC-CCTA	SSFTT-KAN	ST-GGNet
5%	1	97.29	99.36	98.71	99.26	98.64	89.14	98.10	98.18	94.40	**99.47**
	2	98.95	99.42	99.70	99.97	99.79	99.09	99.15	99.87	88.32	**99.99**
	3	91.54	**98.91**	97.85	97.64	96.33	93.00	86.97	97.92	90.18	96.81
	4	99.32	**99.81**	94.16	95.87	99.19	81.50	97.73	89.19	95.52	97.88
	5	97.89	99.99	86.98	99.43	99.58	87.17	**100.00**	96.68	**100.00**	99.94
	6	97.93	99.76	99.54	**100.00**	99.74	99.52	99.23	99.96	**100.00**	**100.00**
	7	96.85	96.82	96.62	99.81	99.49	94.39	96.52	99.08	**100.00**	99.98
	8	93.83	92.80	96.19	97.88	96.23	93.37	94.85	97.75	98.52	**98.55**
	9	99.42	**99.60**	92.45	93.38	94.37	80.13	98.89	81.70	98.78	97.47
	OA	97.61	98.74	98.06	99.11	98.94	93.13	97.86	98.05	93.08	**99.42**
	AA	97.00	98.49	94.60	98.13	98.15	94.83	96.83	95.59	96.19	**98.90**
	k	96.83	98.34	97.43	98.81	98.59	89.08	97.17	97.41	91.11	**99.24**
10%	1	98.25	99.76	99.20	98.81	98.24	97.38	99.48	99.44	94.48	**99.91**
	2	99.43	99.76	99.82	99.98	99.88	99.72	99.95	99.90	88.37	**100.00**
	3	86.34	99.50	99.05	96.54	98.00	96.54	96.72	99.38	90.58	**99.73**
	4	99.05	**99.78**	97.73	97.01	99.72	97.25	98.80	94.28	95.55	97.93
	5	**100.00**	99.97	98.29	99.83	99.84	86.84	**100.00**	98.41	**100.00**	99.75
	6	99.12	99.86	99.96	99.99	99.80	99.65	99.91	**100.00**	**100.00**	**100.00**
	7	97.42	99.99	99.67	99.57	99.53	94.40	99.75	99.50	**100.00**	99.49
	8	94.16	96.67	98.85	97.72	97.73	97.64	99.09	99.22	**99.88**	99.54
	9	**99.86**	99.75	99.49	97.05	99.55	96.71	99.77	86.91	99.07	99.06
	OA	98.04	99.49	99.41	99.15	99.30	97.55	99.55	99.01	93.27	**99.74**
	AA	97.07	99.45	98.52	98.50	99.14	98.15	99.28	97.45	96.44	**99.49**
	k	97.41	99.32	99.21	98.87	99.07	94.30	99.40	98.68	91.35	**99.66**

covers. Specifically, ST-GGNet achieved an OA of 96.88%, AA of 98.01%, and a Kappa coefficient of 96.53%.

3) Results on the Houston: In this experiment, we randomly selected 5% and 10% of samples from each class as training sets for analysis. The test results are shown in Table V. The AA metric, which measures the mean classification accuracy across all classes and is unaffected by class distribution, indicates that when 10% of the dataset is used for training, SSFTT-KAN achieves an AA slightly higher than ours by 0.06. This suggests that with 10% training samples, SSFTT-KAN focuses more on minority classes, performing marginally better in categories such as Class 6 (Water) and Class 9 (Park-Lot1). However, OA and Kappa are more influenced by the majority classes, with OA reflecting overall performance and Kappa emphasizing classification consistency. Our method achieves the highest OA and Kappa scores among all compared methods, indicating superior overall classification performance and consistency. This demonstrates that ST-GGNet handles misclassifications better than SSFTT-KAN. When only 5% of samples are used for training, although SSFTT-KAN and our method achieve the same Kappa score of 96.59%, our OA and AA scores are the highest among all methods, at 96.86% and 97.07%, respectively. This not only indicates that ST-GGNet has lower dependency on training samples compared to SSFTT-KAN, but also our approach maintains better robustness with limited samples.

Fig. 5 illustrates the classification results of the ten models using a 5% training set on the SA dataset. As depicted, models such as SSRN, DBDA, DGPF-RENet, Swin-Transformer, and GSC-CCTA exhibit significant misclassification of Synthetic Grass as Healthy Grass. In contrast, our proposed ST-GGNet demonstrates superior discrimination between these classes. This enhanced performance is attributed to the STM module in ST-GGNet, which leverages spectral feature differences to effectively distinguish spatially similar and adjacent land covers, thereby improving classification accuracy.

4) Results on the WHUHi-Longkou: To effectively demonstrate the robustness and classification performance of our proposed ST-GGNet under limited training samples, we randomly selected 0.2% and 0.5% of samples from each category as training sets for experimental analysis. The results are presented in Table VI. As shown in Table VI, existing SOTA models achieve considerable classification accuracy with a 0.5% training set. However, ST-GGNet achieves the highest OA and Kappa scores among the compared methods, reaching 98.90% and 98.56%, respectively, although its AA score is 1.56% lower than DGPF-RENet. This may be attributed to REM and DGA's suppression of redundant information, enabling precise inter-class differentiation and attaining the highest accuracy across more categories. When using a 0.2% training set, feature selection becomes significantly more challenging. Nonetheless, our model achieves the highest OA, AA, and Kappa scores of 97.74%, 92.48%, and 97.03%, respectively. The AA scores of SOTA models show considerable variation, with ST-GGNet outperforming DGPF-RENet by 5.53% in AA. This indicates that ST-GGNet, through

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TABLE IV
QUANTITATIVE RESULTS OF COMPARATIVE EXPERIMENTS ON SALINAS

Train Sample	Class	Model									
		SSRN	DBDA	Hybrid SN	DGPF-RENet	CS2DT	Swin-Transformer	GTFN	GSC-CCTA	SSFTT-KAN	ST-GGNet
0.5%	1	98.28	99.49	95.03	99.73	**100.00**	**100.00**	99.95	97.56	90.75	99.42
	2	96.00	98.82	99.86	**100.00**	98.36	99.49	99.78	99.98	**100.00**	**100.00**
	3	92.73	96.66	94.09	99.93	99.98	99.29	97.61	98.53	99.39	**99.99**
	4	**97.94**	94.58	95.91	96.64	92.89	86.07	95.61	89.58	96.40	97.69
	5	**99.05**	96.48	86.92	96.18	98.33	94.66	97.26	88.13	97.75	98.71
	6	99.67	99.96	**99.98**	99.35	98.33	99.06	99.64	99.79	96.14	99.90
	7	98.82	96.93	97.24	98.50	**100.00**	**100.00**	99.69	96.03	96.58	99.64
	8	87.81	87.27	79.05	90.41	94.06	**98.10**	91.56	93.16	94.64	92.43
	9	99.14	99.32	95.71	**100.00**	99.75	99.87	99.66	99.73	99.04	**100.00**
	10	94.09	96.77	93.79	99.07	97.12	**99.26**	91.91	96.93	90.93	98.77
	11	88.71	93.40	82.33	99.45	98.30	94.46	98.21	96.72	93.42	**99.83**
	12	98.24	96.47	95.06	98.22	99.77	97.38	**99.84**	96.15	94.27	98.96
	13	96.35	**99.70**	96.33	99.24	95.96	90.67	90.35	96.47	94.30	97.70
	14	94.58	95.44	96.29	96.16	92.80	91.63	**98.22**	92.75	69.23	93.90
	15	70.50	78.09	78.27	92.59	89.55	**99.22**	88.03	96.00	83.84	92.11
	16	98.05	99.16	98.61	99.26	**100.00**	**100.00**	96.83	94.22	66.20	99.14
	OA	88.32	91.19	89.20	96.33	96.35	96.48	95.36	95.95	92.81	**96.88**
	AA	94.37	95.53	92.78	97.80	97.20	96.07	96.51	95.73	91.43	**98.01**
	k	87.10	90.20	87.93	95.92	95.94	96.09	94.84	95.49	92.03	**96.53**
1%	1	98.69	99.95	**100.00**	99.93	**100.00**	**100.00**	**100.00**	99.39	90.71	99.74
	2	99.86	99.41	99.38	99.99	99.45	99.87	**100.00**	99.99	**100.00**	**100.00**
	3	95.71	98.90	96.00	**100.00**	99.98	**100.00**	99.80	99.84	99.39	99.95
	4	98.81	95.61	97.89	98.76	94.89	80.45	**99.49**	95.57	91.68	98.64
	5	98.91	**99.35**	95.76	98.61	99.33	95.30	96.19	97.08	65.74	97.87
	6	**99.99**	99.98	99.97	99.86	99.76	94.56	99.95	99.21	96.18	99.81
	7	99.90	99.85	99.49	99.94	**100.00**	**100.00**	99.92	99.57	96.31	99.79
	8	88.43	96.05	85.49	95.87	95.50	96.80	96.66	**99.01**	94.45	96.19
	9	99.60	99.40	98.77	**100.00**	99.87	99.44	99.89	99.85	99.06	**100.00**
	10	97.62	98.22	96.69	**99.68**	98.67	98.82	98.49	98.70	93.78	99.62
	11	95.29	97.34	95.83	**99.73**	99.53	98.21	99.62	99.06	94.24	99.11
	12	94.24	99.42	98.88	99.80	**99.96**	98.24	99.69	99.14	94.24	99.00
	13	97.10	**99.85**	97.00	98.87	99.70	85.63	98.79	94.12	95.04	98.80
	14	98.02	97.47	**98.17**	94.89	95.76	90.49	96.51	95.16	88.60	97.07
	15	87.32	83.12	77.93	96.18	93.49	92.99	97.19	**98.98**	88.52	98.51
	16	98.94	99.95	99.25	99.60	**100.00**	**100.00**	97.43	79.19	71.52	99.68
	OA	94.32	95.37	92.21	98.34	97.76	98.13	98.41	98.25	93.92	**98.66**
	AA	96.78	97.74	96.03	98.86	98.49	96.90	98.73	97.12	93.09	**98.99**
	k	93.67	94.87	91.33	98.16	97.50	97.72	98.23	98.05	93.27	**98.51**

its superior nonlinear processing capabilities, mitigates information redundancy and the curse of dimensionality under minimal training samples, offering advantages in feature learning with limited data.

Fig. 6 presents the classification visualization for the WHUHi-Longkou dataset using 0.2% samples as the training set. While most models exhibit misclassifications in continuous regions of cotton, sesame, and broad-leaf soybean, as well as confusion between mixed weed and roads and houses, ST-GGNet stands out. Specifically, SSRN and SSFTT-KAN predominantly misclassify sesame as broad-leaf soybean. In contrast, our model maintains accurate classifications in these regions, including at the boundary between mixed weeds and roads and houses. This demonstrates ST-GGNet's strong generalization capabilities, even with a very small proportion of training samples.

E. Module Effectiveness Experiment

1) Experimental Effectiveness of B-Spline Function Order in the KAN Layer: In this article, B-Splines were introduced to improve the nonlinear transformation power of the KAN layer for efficiently processing high-dimensional data in HSI. The influence of B-Spline orders on model performance differs. Lower degree splines, such as Linear and Quadratic, help reduce overfitting but struggle to capture complex nonlinear relationships. In contrast, higher-degree splines like Cubic, Quartic, and Quintic can better represent intricate nonlinearities, though they come with a higher risk of overfitting.

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TABLE V
QUANTITATIVE RESULTS OF COMPARATIVE EXPERIMENTS ON HOUSTON

Train Sample	Class	Model									
		SSRN	DBDA	Hybrid SN	DGPF-RENet	CS2DT	Swin-Transformer	GTFN	GSC-CCTA	SSFTT-KAN	Ours
5%	1	94.30	95.19	86.78	95.15	95.60	92.17	86.30	97.80	**97.99**	94.44
	2	98.67	99.00	86.36	97.87	98.35	85.92	**99.57**	98.50	98.58	99.24
	3	99.57	99.93	98.61	99.56	**100.00**	98.62	90.55	89.34	99.10	99.55
	4	98.29	98.44	91.88	98.47	95.88	91.48	**99.92**	95.58	99.16	97.63
	5	97.07	99.45	**100.00**	**100.00**	99.97	99.49	2.47	98.89	98.23	**100.00**
	6	97.88	99.79	99.63	96.44	97.35	99.62	73.61	85.75	87.14	**100.00**
	7	96.27	90.74	82.76	96.19	97.23	83.70	63.49	95.97	**97.44**	96.10
	8	95.70	98.08	97.72	91.27	**99.38**	98.18	36.76	96.95	94.37	91.71
	9	93.37	87.32	88.46	90.12	95.93	86.28	66.58	**96.62**	91.31	93.78
	10	87.94	85.38	95.95	98.56	93.81	96.11	79.58	**99.99**	99.83	99.06
	11	93.50	95.51	96.78	98.49	92.62	96.34	72.86	**99.77**	96.87	98.04
	12	90.75	95.72	93.15	96.61	95.43	94.22	38.12	**98.85**	94.83	95.73
	13	97.99	98.22	**99.71**	88.63	95.21	95.54	96.95	98.08	92.43	90.81
	14	98.42	99.69	99.27	**100.00**	**100.00**	97.84	99.39	**100.00**	**100.00**	**100.00**
	15	98.90	98.50	93.78	99.82	97.60	89.10	75.19	49.98	**100.00**	**100.00**
	OA	94.97	94.71	92.68	96.45	96.62	92.69	75.19	95.19	96.83	**96.86**
	AA	95.91	96.06	92.36	96.48	96.96	92.54	72.50	93.47	96.48	**97.07**
	k	94.57	94.28	92.08	96.16	96.34	92.10	73.13	94.79	**96.59**	**96.59**
10%	1	95.88	97.94	98.27	98.68	**99.69**	97.98	90.80	98.29	99.65	99.20
	2	98.96	98.77	96.05	99.34	99.67	97.50	96.54	98.35	**100.00**	99.58
	3	**100.00**	99.96	97.94	99.68	10.00	98.09	99.13	99.20	99.84	**100.00**
	4	99.55	99.22	93.90	99.16	99.02	97.26	96.35	95.37	**100.00**	99.38
	5	99.55	99.57	99.29	**100.00**	99.96	99.29	99.92	99.01	98.14	**100.00**
	6	**100.00**	**100.00**	91.26	98.76	**100.00**	93.38	94.41	77.60	**100.00**	97.94
	7	97.43	**99.16**	90.86	98.04	97.98	89.82	82.64	96.09	98.44	98.77
	8	**99.26**	98.66	98.43	97.42	98.52	98.65	54.22	97.36	97.88	98.66
	9	**98.09**	96.06	92.44	97.24	95.61	92.68	82.42	96.95	97.63	96.36
	10	96.72	95.35	99.18	99.78	96.46	99.73	75.47	99.93	**100.00**	**100.00**
	11	98.27	97.36	99.46	99.71	94.76	**100.00**	88.14	99.27	98.58	**100.00**
	12	94.95	96.09	97.93	98.46	98.52	**99.09**	58.64	99.05	98.40	98.83
	13	97.82	99.04	99.23	95.23	99.35	99.02	63.66	97.46	**100.00**	99.01
	14	**100.00**	99.83	**100.00**	**100.00**	**100.00**	**100.00**	99.29	**100.00**	**100.00**	**100.00**
	15	99.07	98.96	98.31	99.98	98.84	97.99	99.39	79.44	**100.00**	**100.00**
	OA	98.02	98.05	96.71	98.80	98.21	97.24	84.21	96.81	99.06	**99.16**
	AA	98.37	98.40	96.75	98.77	98.56	97.36	84.40	95.56	**99.24**	99.18
	k	97.86	97.90	96.45	98.70	98.06	97.01	82.93	96.54	98.98	**99.09**

To identify the optimal spline degree, we conducted a comparative analysis of classification performance across Linear, Quadratic, Cubic, Quartic, and Quintic splines, categorizing Linear and Quadratic as low-degree and Cubic, Quartic, and Quintic as high degree. The experimental results are presented in Fig. 7.

As illustrated, the Cubic spline achieves OA and Kappa coefficients similar to the Quintic spline, while significantly surpassing it in AA. This indicates that Cubic B-splines strike the best balance between mitigating overfitting and effectively capturing complex nonlinear relationships, resulting in optimal performance for the ST-GGN model. Therefore, we choose Cubic B-splines for the KAN layer to maximize the model's effectiveness.

2) Experimental Effectiveness of Parameter θ in the STM Module: This section investigates the impact of the hyperparameter θ within STM on model performance. The setting of θ directly influences the model's reliance on gradient information and the overall feature extraction approach. We conducted experiments with θ values ranging from 0 to 1 to determine the optimal parameter value. Using OA, AA, and Kappa as evaluation metrics, we tested values of 0.0, 0.2, 0.5, 0.7, and 1.0 on the 0.2% Salinas dataset. The evaluation results are depicted in the following Fig. 8. Our experimental results indicate that when θ is set to 0.0 and 0.2, the Kappa metric yields nearly identical values. However, compared to $\theta = 0.0$, both AA and OA metrics attain higher values at $\theta = 0.2$. Consequently, the best performance is observed at $\theta = 0.2$.

Therefore, after a comprehensive evaluation, we selected $\theta = 0.2$ as the hyperparameter value for this study.

3) Experimental Effectiveness of BGO: To validate the effectiveness of BGO, this section conducts experiments

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TABLE VI
QUANTITATIVE RESULTS OF COMPARATIVE EXPERIMENTS ON WHUHI-LONGKOU

Train Sample	Class	Model									
		SSRN	DBDA	Hybrid SN	DGPF-RENet	CS2DT	Swin-Transformer	GTFN	GSC-CCTA	SSFTT-KAN	Ours
0.2%	1	93.48	98.33	97.52	**99.65**	93.60	97.20	97.78	97.88	91.04	99.56
	2	50.23	80.17	74.76	88.43	83.16	97.66	64.79	87.65	85.50	**98.01**
	3	65.39	84.53	88.43	92.66	96.53	**96.92**	94.02	73.47	11.42	94.99
	4	87.09	94.72	91.86	97.98	92.98	97.93	97.62	98.13	90.94	**98.86**
	5	35.42	71.37	84.84	71.64	**91.56**	83.48	86.47	86.68	69.86	82.40
	6	80.64	**98.55**	98.26	95.43	97.20	97.50	92.76	91.50	63.82	97.95
	7	99.16	99.78	99.54	**99.84**	98.42	98.70	98.88	99.45	96.62	99.78
	8	70.08	83.61	81.34	76.96	85.38	58.29	77.14	66.21	77.99	**86.34**
	9	49.51	91.33	**93.99**	59.98	90.03	52.91	74.04	55.85	15.52	74.48
	OA	87.28	95.21	94.29	96.02	94.49	95.12	94.69	94.92	87.04	**97.74**
	AA	70.11	89.16	90.06	86.95	92.10	86.78	85.94	84.09	66.97	**92.48**
	k	83.23	93.69	92.43	94.74	92.69	93.6	93.00	93.29	83.05	**97.03**
0.5%	1	98.8	99.63	99.56	99.70	99.21	98.62	99.56	99.27	91.43	**99.75**
	2	75.19	93.98	89.47	98.86	93.25	**99.45**	93.69	98.25	92.68	98.37
	3	91.04	93.06	97.7	**99.94**	98.95	95.41	92.97	88.52	92.48	98.79
	4	95.93	98.39	98.67	98.24	98.75	99.28	98.20	99.04	92.48	**99.32**
	5	76.44	90.18	96.11	**99.44**	94.98	90.05	89.98	95.56	90.95	95.18
	6	98.71	99.67	**99.68**	99.56	99.62	97.32	98.77	97.32	95.64	99.29
	7	99.94	99.96	**99.97**	98.83	99.88	99.56	99.93	99.57	97.85	99.85
	8	91.9	**96.73**	93.55	93.57	94.62	80.12	93.79	79.92	96.25	91.57
	9	89.6	88.17	95.93	**96.96**	95.82	75.75	90.58	79.51	90.76	88.88
	OA	95.88	98.22	98.49	98.64	98.73	97.61	98.25	97.73	94.31	**98.90**
	AA	90.84	95.53	96.74	**98.34**	97.23	93.56	95.27	93.00	93.39	96.78
	k	94.58	97.67	98.02	98.21	98.33	96.86	97.70	97.01	92.64	**98.56**

comparing four widely used optimizers in the HSI classification domain.

1) SGD optimizer with a momentum factor of 0.9 and enabled Nesterov acceleration.
2) Adaptive gradient algorithm (Adagrad) optimizer with a numerical stability constant $\varepsilon = 10^{-8}$.
3) Adaptive moment estimation (Adam) optimizer with first moment decay rate $\beta_1 = 0.9$ and second moment decay rate $\beta_2 = 0.999$.
4) Root mean square propagation (RMSProp) optimizer with an exponential decay rate $\rho = 0.9$.

All other experimental parameters followed the settings outlined in Sections V-A and V-B. These comparisons further demonstrate the effectiveness of BGO in HSI classification tasks. The experimental results are presented in Table VII. Table VII indicates that training the ST-GGN model using BGO achieves more desirable classification outcomes compared to other classical optimizers. Notably, the RMSProp optimizer can be considered a simplified version of BGO without the budding growth mechanism. Comparative analysis between these two optimizers reveals that although using BGO results in a slightly longer training time than RMSProp by approximately 0.36 s, the model's final performance significantly improves. Additionally, the variances of OA, AA, and Kappa coefficients are smaller, indicating a more stable training process. This suggests that BGO, in Table VII, through its budding growth mechanism, effectively promotes stable convergence and enhances the model's ability to find superior solutions when handling high-dimensional HSI data, thereby achieving more accurate classification results.

To facilitate a more direct observation of the convergence performance of different optimizers toward the optimal solution, we fix the values of all parameters except for two weights in a convolutional layer and observe the convergence behavior of different optimizers on these two parameters during training. The results are shown in Fig. 9, where the red origin represents the original weight values, and the closer the training path's endpoint is to the yellow region, the lower the loss, indicating more accurate classification. As seen in Fig. 9, the paths of SGD and Adagrad are relatively short, with no stable descent on the loss surface. Adam shows a good downward trend but still fails to fully converge at the end of training. RMSprop experiences significant fluctuations during convergence, indicating instability. In contrast, BGO, using a budding growth mechanism, effectively explores better solutions and converges to a lower loss. Moreover, the perturbation factor does not cause significant fluctuations in convergence, and BGO maintains stable convergence while preserving exploration capacity in the later stages of training compared to the other four optimizers.

F. Ablation Experiments

To validate the effectiveness of the proposed method, we implemented various structural configurations and conducted ablation studies to evaluate the contribution of each

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TABLE VII
EXPERIMENTAL RESULTS DEMONSTRATING OPTIMIZER EFFECTIVENESS

Optimizer	OA	AA	Kappa	Train time(s)	Test time(s)
SGD	51.36±3.92	34.22±2.73	43.50±4.78	**11.50±0.44**	**12.52±1.11**
Adagrad	75.75±2.15	58.84±1.857	72.68 ±2.34	12.18±0.43	12.73±0.29
Adam	88.31±2.56	89.19±3.19	86.99±2.82	14.61±0.41	12.56±0.36
RMSprop	88.95±2.89	90.78±4.06	89.14±3.20	12.73±0.28	12.91±0.34
BGO	**91.96±1.94**	**92.21±2.20**	**91.05±2.20**	13.09±0.27	12.83 ± 0.26

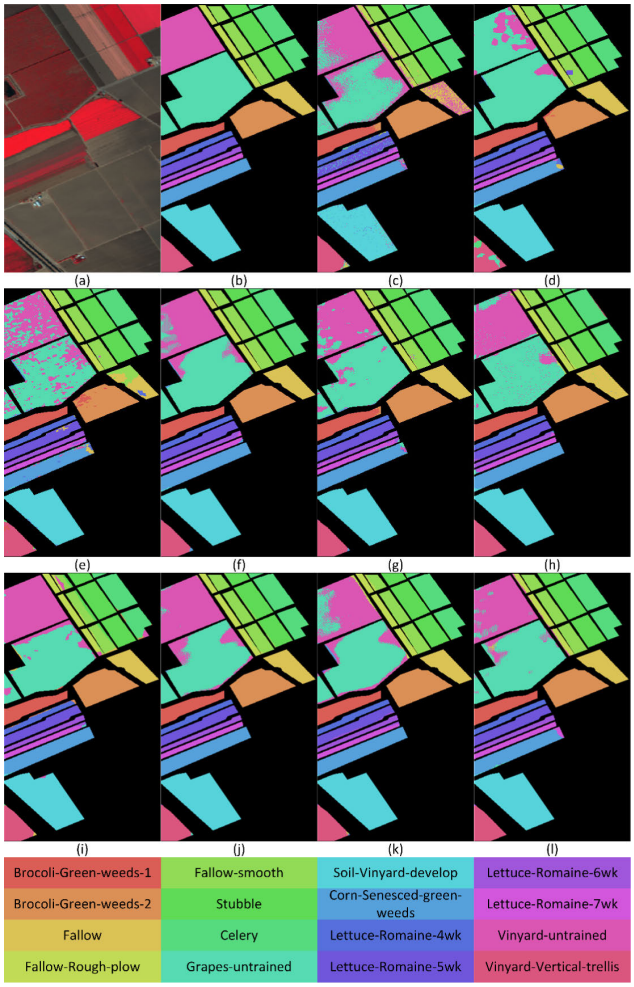


Fig. 4. Visualization results of comparative experiments on Salinas. (a) Input. (b) GT. (c) SSRN. (d) DBDA. (e) HybridSN. (f) DGPF-RENet. (g) CS2DT. (h) Swin-Transformer. (i) GTFN. (j) GSC-CCTA. (k) SSFTT-KAN. (l) ST-G-GNet.

submodule within ST-GGNet to the overall performance. The ablation experiments involved systematically removing STM, GGA, and BGO using a controlled variable approach. Specifically: Group 1 represents ST-GGNet, Group 2 removes ST-GGNet 's BGO, Group 3 removes ST-GGNet 's GGA, Group 4 removes ST-GGNet 's STM, Group 5 removes ST-GGNet 's BGO and GGA, and Group 6 removes ST-GGNet 's BGO and STM. Group 7 removed GGA and STM from ST-GGNet, and Group 8 removed all three modules of BGO, GGA, and STM from ST-GGNet. The experimental results are given in Table VIII. Certain critical results are shown in Fig. 10. The analysis of Table VIII and Fig. 10 reveals the following.

1) The comparison of Groups 1 and 2, removal of BGO led to declines in all metrics relative to the complete ST-GGNet (OA: −2.01%, AA: −1.43%, Kappa: −1.91%). This indicates that the budding growth mechanism is essential for the model's exploratory capability and stable convergence to an optimal solution. Without the BGO module, the integration benefits of STM and GGA are diminished, limiting overall performance.
2) The comparison of Groups 1 and 3, removal of GGA, resulted in the most significant declines in all metrics relative to the complete ST-GGNet (OA: −4.42%; AA: −9.6%; Kappa: −5.31%). This performance drop indicates that when the model is restricted to processing only local information due to the absence of GGA, it fails to capture dependencies between distant nodes, leading to suboptimal performance. It also confirms that the nonlinear transformation in the KAN layer facilitates deeper interactive fusion of HSI's nonlinear data in the Glance and Gaze modules.
3) The comparison of Groups 1 and 4, removal of STM, resulted in significant declines in all metrics relative to the complete ST-GGNet (OA: −4.09%; AA: −5.24%; Kappa: −3.83%). This performance drop indicates that without STM, the model fails to capture multiscale spatial features and spectral information in HSI, leading to lower utilization of HSI data and exacerbated performance degradation under small-sample conditions. It further demonstrates that STM's integration of multiscale central-difference convolution effectively captures fine details and texture features in HSI images, while its graph structure modeling of the output features fully leverages spectral information, thereby enhancing HSI data representation.
4) The comparison of Groups 2 and 6, the integration of STM and GGA achieved significantly better results than using GGA alone (OA: +2.47%; AA: +5.33%; Kappa: +2.38%), representing a marked improvement. This not only demonstrates the substantial enhancement of the GGA module by STM but also indicates that the nonlinear transformation in the KAN layer effectively processes HSI information modeled in nonEuclidean space. Moreover, when faced with the abstract and extensive features generated by STM, the GGA mechanism adaptively reinforces critical information while suppressing irrelevant data, thereby fully exploiting the

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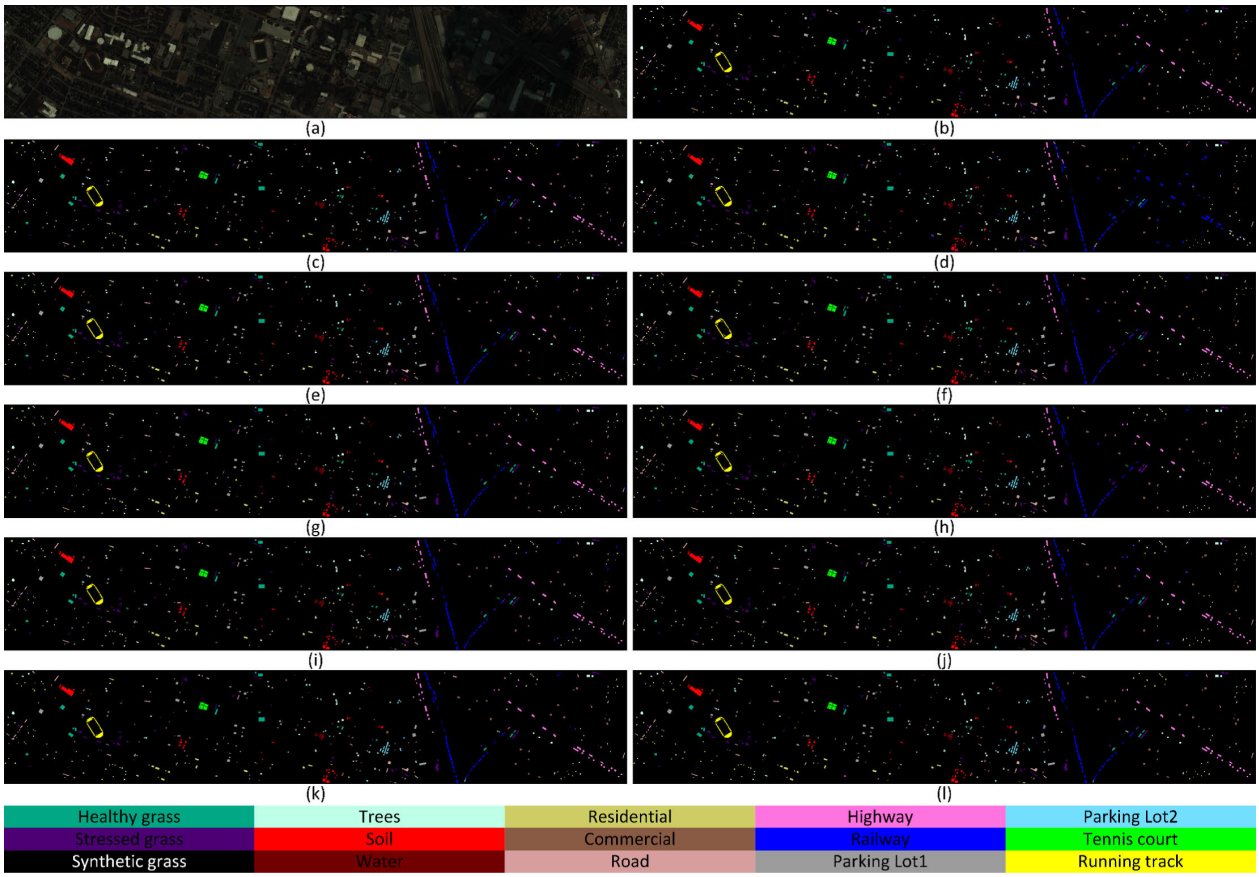


Fig. 5. Visualization results of comparative experiments on Houston. (a) Input. (b) GT. (c) SSRN. (d) DBDA. (e) HybridSN. (f) DGPF-RENet. (g) CS2DT. (h) Swin-Transformer. (i) GTFN. (j) GSC-CCTA. (k) SSFTT-KAN. (l) ST-GGNet.

TABLE VIII
ABLATION EXPERIMENT RESULTS

Method	Group	STM	GGA	BGO	OA	AA	Kappa
ST-GGN	**1**	√	√	√	**91.96**	**92.21**	**91.05**
	2	√	√	×	89.95	90.78	89.14
	3	√	×	√	87.54	82.61	85.74
	4	×	√	√	87.87	86.97	87.22
	5	√	×	×	86.17	81.44	82.77
	6	×	√	×	87.48	85.45	86.76
	7	×	×	√	84.57	83.75	84.68
	8	×	×	×	83.25	83.22	83.91

rich details and global modeling captured by STM. These findings further validate the effectiveness of integrating STM and GGA to enhance overall model performance.

5) The comparison of Groups 5 and 8, the results show that, relative to the baseline, the incorporation of GGA alone yields the most significant improvement in model accuracy (OA: + 4.23%; AA: +2.23%; Kappa: +2.85%). This enhancement is primarily attributed to the nonlinear transformation in the KAN layer, which facilitates the extraction of nonlinear abstract relationships within HSI. Furthermore, the KAN-based GGA mechanism leverages a visual attention strategy to reinforce the understanding of HSI feature maps. This not only demonstrates the strong portability of GGA but also confirms the successful integration of the KAN layer in processing HSI data.
6) The comparison of Groups 6 and 8 indicates that, relative to the baseline, the incorporation of only STM yields the least improvement in model performance (OA: +2.92%; AA: −1.78%; Kappa: −1.14%), with the AA and Kappa metrics even falling below those of the baseline method. Upon analysis, we attribute this to the high-dimensionality and abstraction of the features

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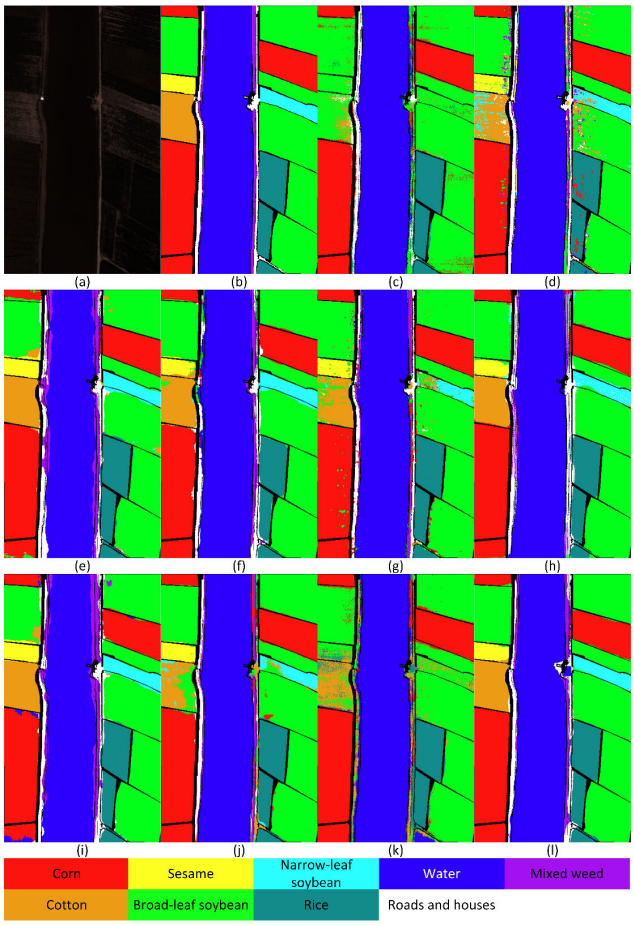


Fig. 6. Visualization results of comparative experiments on WHU-Hi Longkou. (a) Input. (b) GT. (c) SSRN. (d) DBDA. (e) HybridSN. (f) DGPF-RENet. (g) CS2DT. (h) Swin-Transformer. (i) GTFN. (j) GSC-C-CTA. (k) SSFTT-KAN. (l) ST-GGNet.

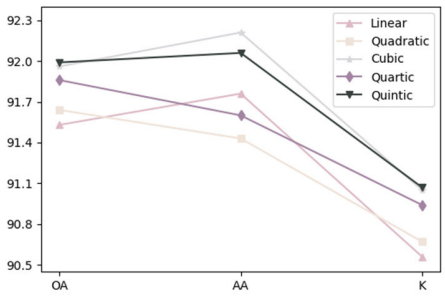


Fig. 7. Comparative experiments were conducted using Linear, Quadratic, Cubic, Quartic, and Quintic splines to evaluate the model performance with different orders in the KAN layer.

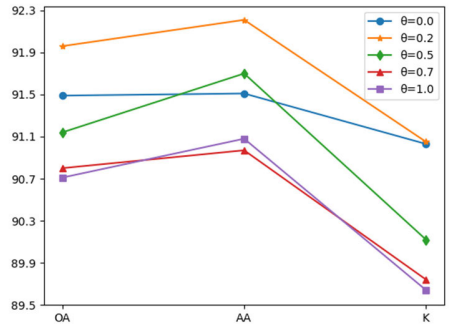


Fig. 8. Value of OA, AA, K when is set to 0.0, 0.2, 0.5, and 0.7.

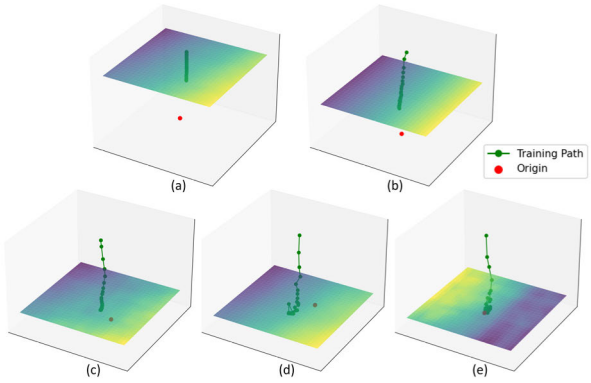


Fig. 9. (a) SGD. (b) Adagrad. (c) Adam. (d) RMSprop. (e) BGO.

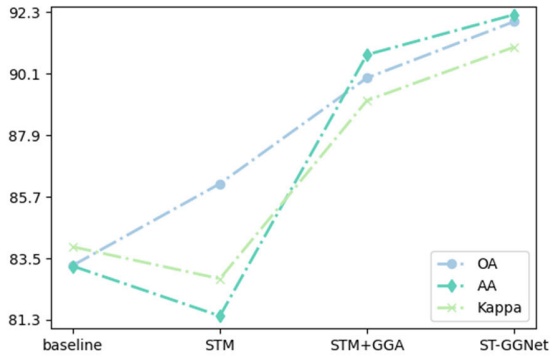


Fig. 10. Results of the ablation experiments are presented. The "baseline" corresponds to Group 8 in Table VIII; "STM" refers to ST-GGNet without the GGA and BGO modules, corresponding to Group 5 in Table VIII; "STM + GGA" represents ST-GGNet with the BGO module, corresponding to Group 2 in Table VIII; "ST-GGNet" corresponds to Group 1 in Table VIII.

produced by STM's multiscale CDC method combined with graph structure modeling, which challenges the model's ability to identify key features without an appropriate supplementary layer to process these feature maps. These findings suggest that STM has limited portability and is difficult to directly integrate into other networks.

7) The comparison of Groups 7 and 8 indicates that BGO alone still yields performance gains (OA: +1.32%; AA: +0.53%; Kappa: +0.77%). BGO enhances path diversity and mitigates repetitive optimization along similar directions, facilitating exploration of a broader parameter space. This leads to clearer inter-class feature separation and reduced confusion, further validating BGO's effectiveness in boosting model performance.

VI. Results and Discussion

Experimental results in Section V-D demonstrate that ST-GGNet consistently outperforms other SOTA methods under low-sample conditions. This chapter first analyzes the performance of ST-GGNet, then compares its computational complexity with other SOTA models, and finally discusses its potential limitations.

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A. Comprehensive Summary and Significance of Experimental Results

The four datasets vary in sensor platforms, spectral ranges, spatial resolutions, and land-cover compositions, posing different challenges for feature extraction and generalization. These variations provide a valuable basis for assessing ST-GGNet's adaptability to diverse scenarios. To better interpret the experimental results and explore their significance, we conduct an integrated analysis across all datasets. This helps elucidate the impact of dataset-specific differences on model performance and clearly demonstrates the advantages and practical value of ST-GGNet. This subsection is different from Section V-D, which emphasizes comparison with other SOTA models; the following discussion focuses on dataset-wise evaluation of ST-GGNet.

1) Performance and Significance on Pavia University: This dataset focuses on an urban environment containing many similarly appearing rooftops and bare soil, alongside a complex background due to uneven land-cover distributions. Such conditions can lead to small isolated regions and spectrally similar adjacent categories. Despite relying on only 5% labeled samples, ST-GGNet maintains an OA of 99.42%, indicating high robustness to boundary noise and small-sample constraints. The STM module's graph-based spectral similarity structure and CDC multiscale convolution enhance the extraction of spectral and textural features, reducing misclassifications caused by "spectrally identical but different classes" or "same class but varying spectra." These findings demonstrate that ST-GGNet achieves high-performance classification with limited training samples in urban HSI classification tasks. Furthermore, it exhibits robust performance in complex backgrounds and in the presence of boundary noise.

2) Performance and Significance on Salinas: The Salinas dataset, covering an agricultural scene, features distinct inter-class differences and relatively uniform spectral patterns in planting areas. Despite having 16 categories, including spectrally similar vegetable and vineyard classes, ST-GGNet achieves an AA of 98.01% with only 0.5% training data. This highlights the GGA module's effectiveness in mitigating confusion caused by overlapping spectral bands. The "Glance" operation captures global spectral relationships, while "Gaze" focuses on local differences, enhancing class separability. These findings demonstrate that STGGNet achieves precise classification even among spectrally and spatially similar land cover features in agricultural environments. This highlights its capability for fine-grained spectral–spatial discrimination and its significant application potential in accurate field classification tasks.

3) Performance and Significance on Houston: This large-scale urban dataset contains 15 diverse classes, including both artificial and natural surfaces of the earth. With a spatial resolution of 2.5 m, it introduces notable background noise and mixed pixels. ST-GGNet outperforms other methods across OA, AA, and Kappa metrics under both 5% and 10% training rates, especially in distinguishing hard-to-separate classes like healthy and synthetic grass. This advantage stems from STM's spectral similarity graph construction, which enhances the model's ability to separate classes with subtle spectral differences. Even within this highly challenging dataset, ST-GGNet demonstrates precise classification of mixed pixels with indistinct edges, relying on subtle spectral differences. This further validates its robustness and generalization capabilities in real-world applications.

4) Performance and Significance on WHUHi-LongKou: This UAV-based dataset features higher spatial resolution, leading to complex textures and more pronounced spectral similarity across different land-cover types. Under an extremely low labeling rate of 0.2%, ST-GGNet achieves 97.74% OA and 97.03% Kappa. This is enabled by GGA's dual attention to global and local contexts and BGO's improved convergence, which together mitigate overfitting and local optima. High Kappa values on cross-validation further confirm the model's stability. In multiclass field scenes with complex textures, ST-GGNet achieves high accuracy using only 0.2% of the trainable samples. This demonstrates its effective long-range modeling capability for targeted key feature attention and redundant information filtering of complex texture features. Consequently, ST-GGNet exhibits superior feature processing for the large amount of captured feature information in low-sample classification scenarios.

B. Computational Complexity

To more comprehensively evaluate ST-GGNet, we use only 0.5% of the Salinas dataset as training samples to compare the computational complexity with other SOTA models, including parameters (Params), floating-point operations (FLOPs), training time, and testing time. To ensure fairness, the same training configurations specified in Section V-A are adopted. The experimental results are shown in Table IX. Although ST-GGNet has the highest parameter count, it achieves lower FLOPs and training time than six SOTA methods, and faster test time than seven, demonstrating an effective balance between accuracy and computational cost. While it is not the most lightweight model, its efficiency under low-sample conditions is notable.

Furthermore, to evaluate the inference speed and resource constraints of ST-GGNet from a practical deployment perspective, we conducted deployment and testing using only a 2.5 GHz 12-core Platinum 8255C CPU without GPU acceleration, under the hardware configuration described in Section V-A. The results are summarized in Table X.

As shown in Table X, the accuracy under CPU-only execution remains nearly identical to that reported for Group 1 in Table VIII with GPU acceleration. Notably, ST-GGNet achieves an inference speed of 16.2 ms per HSI image, corresponding to 61.65 frames per second (FPS), even without GPU support. This demonstrates its strong applicability and practical value in resource-constrained environments, particularly for real-time applications such as high-performance edge servers.

C. Scalability Evaluation on Larger Datasets

In Section V-D, we have extensively evaluated ST-GGNet on several widely used HSI datasets. While these datasets are reasonably challenging, they involve a limited number of

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TABLE IX
COMPUTATIONAL COMPLEXITY OF MODELS, INCLUDING PARAMS, FLOPS, TRAIN TIME, AND TEST TIME

Methods	Params (M)	FLOPs (M)	Train time(s)	Test time(s)
SSRN	0.37	161.60	45.38	20.75
DBDA	0.39	110.17	34.15	28.11
Hybrid-SN	0.91	1.83	5.65	2.27
DGPF-RENet	3.49	7.00	17.31	10.24
CS2DT	1.99	250.00	29.39	45.22
Swin-Transformer	0.09	2,282.05	26.84	16.80
GTFN	0.29	1,104.89	26.25	25.39
GSC-CCTA	0.35	0.31	18.34	14.16
SSFTT-KAN	0.34	35.99	14.14	17.15
ST-GGNet	3.97	8.95	18.10	12.71

TABLE X
PERFORMANCE TEST WITH CPU ONLY

OA	AA	Kappa	Inference time(s/per)	FPS
91.32	91.69	90.21	16.2ms	61.65

classes—typically around a dozen—making it insufficient to fully validate ST-GGNet's scalability and robustness in large-scale, multiclass scenarios. Larger datasets with more diverse categories impose greater demands on a model's generalization ability and classification robustness in complex environments.

To this end, we further evaluate ST-GGNet on the WHU-Hi-Honghu dataset, which contains 22 land-cover classes. To highlight ST-GGNet's effectiveness under limited training data, only 1% of labeled samples per class are used for training. Each experiment is repeated ten times, and the average results are reported. The detailed experimental settings are described in Section V-A. The category names, the number of training samples per class, and the corresponding classification results are summarized in Table XI.

As shown in Table XI, ST-GGNet achieves outstanding classification performance on WHU-Hi-Honghu. Most categories reach or exceed 90% accuracy, demonstrating strong multiclass recognition capability and robustness on large-scale datasets. Furthermore, a Kappa coefficient of 97% indicates the model's excellent discriminative power. Notably, ST-GGNet achieves over 97% OA while maintaining high per-class precision across 22 heterogeneous land-cover types, confirming its scalability and generalization capacity for large-scale HSI classification.

TABLE XI
CATEGORIES OF THE WHU-HI-HONGHU DATASET, THE NUMBER OF TRAINING SAMPLES, AND ACCURACY ACHIEVED BY ST-GGNET

Class	Samples	Accuracy(%)
Red roof	140	98.22
Road	35	81.91
Bare soil	218	97.83
Cotton	1632	99.92
Cotton firewood	62	95.71
Rape	445	99.78
Chinese cabbage	241	94.82
Pakchoi	40	92.45
Cabbage	108	99.09
Tuber mustard	123	92.39
Brassica parachinensis	110	95.75
Brassica chinensis	89	86.65
Small Brassica chinensis	225	96.13
Lactuca sativa	73	90.33
Celtuce	10	94.15
Film covered lettuce	72	97.86
Romaine lettuce	30	99.43
Carrot	32	90.58
White radish	87	95.08
Garlic sprou	34	95.19
Broad bean	13	95.59
Tree	40	98.58
OA	-	97.69
AA	-	94.88
Kappa	-	97.07

D. Potential Weakness

This study aims to construct a deep learning network with low sample dependency for HSI classification in data-limited scenarios. The STM module, through multiscale convolutions combined with CDC and graph-based spectral modeling, captures deep texture and spectral features, reducing sample dependency. Although this enhances feature discrimination, it also increases feature complexity. As shown in Table VIII, without the GGA attention mechanism that effectively handles nonlinear transformations, STM maybe negatively affect the baseline performance. This highlights both the success of integrating STM with GGA and the limited portability of STM. According to our analysis, the STM integrates multiscale convolutions with CDC and graph-based spectral modeling to capture intricate texture and spectral features, thereby lowering sample dependency. Although this improves feature discrimination, it also heightens feature complexity. Fig. 11 substantiates this view; in Fig. 11(a), the STM feature map exhibits stronger overall intensity, indicating that it encodes richer information and consequently increases the baseline model's computational burden. Consequently, leveraging an effective attention strategy (such as GGA) is essential to emphasize pivotal features, as shown in Fig. 11(b).

To address this, future work will introduce an interface mechanism that dynamically regulates information flow, such as neighborhood filtering or adaptive weighting within STM, to simplify the feature maps generated by STM. We hope that this mechanism filters the rich features extracted by STM,

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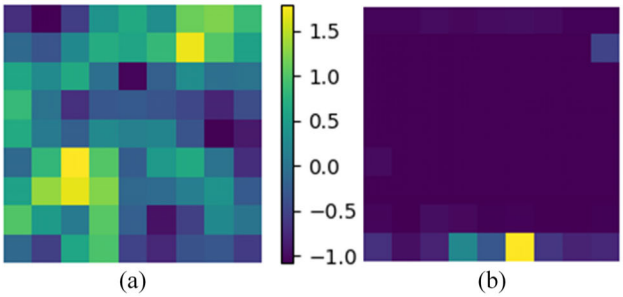


Fig. 11. Feature maps from the same channel output. (a) Feature map of STM outputs. (b) Feature map of GGA outputs. The central vertical bar represents the intensity scale, where brighter regions indicate stronger response magnitudes.

improving the processing efficiency of subsequent layers and reducing reliance on GGA. Additionally,

Table IX also reveals that the model's large parameter count leads to a higher dependence on storage capacity. Therefore, a key direction for future research will be the integration of lightweight modules, enabling us to achieve comparable or even superior performance with reduced resource demands.

It should be noted that real-world HSI classification typically involves additional preprocessing procedures, such as sub-pixel localization, atmospheric correction, band compression, and superpixel fusion. These steps may introduce additional computational overhead. However, the design and integration of these modules for real-time deployment fall outside the scope of this study. Future research will focus on model compression, lightweight architecture design, and modular integration strategies to develop an end-to-end HSI processing framework suitable for real-time applications.

VII. Conclusion and Outlook

In this article, we proposed the ST-GGNet to address the core challenges in HSI classification, including limited sample availability, inadequate long-range dependency modeling, and the risk of convergence to local optima. By incorporating the STM, GGA, and BGO, ST-GGNet enhances spectral–spatial feature extraction, selectively focuses on informative components, and achieves more stable and accurate training. Extensive experiments on four public benchmark datasets demonstrate that ST-GGNet outperforms the nine compared SOTA models, exhibiting superior performance under limited training data. Although the classification results are promising, a significant limitation remains in the poor portability of STM. Future work will research a dynamic interface mechanism that adjusts information flow to enhance the adaptability of STM output features to downstream processing, thus improving portability.

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